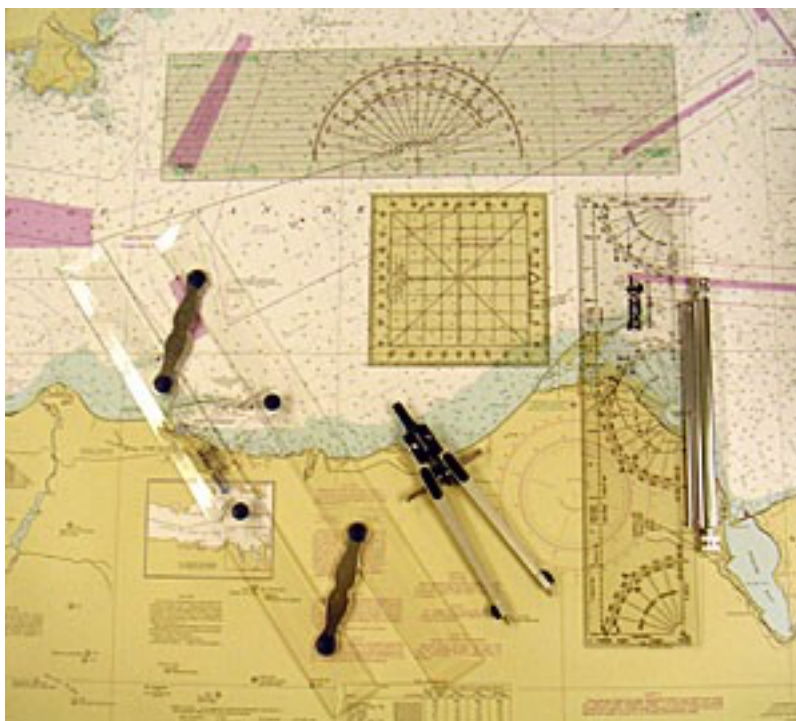


!"#\$%&'"(%)&*+"%, "- .)' /0.1"(\$.&'*" ., 2"3.* /0"41%' '/, 5"
6%&""\$+"7+0&+. ' /%, .1"3%. '+&"



Gary C. Kessler

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<https://scuba.garykessler.net>

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14 Febuary 2023

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Chart No. 1: Nautical Chart Symbols Abbreviations and Terms used on Paper and Electronic Navigational Charts

U.S. Coast Pilot®

Light List (Volume I: Atlantic Coast Volume III: Atlantic and Gulf Coasts)

Notice to Mariners Local Notice to Mariners

U.S. Aids to Navigation

Navigation Rules and Regulations Handbook

60 D ST

Introduction

Suggestions, corrections, and/or any other comments about this manual are welcome.

Part 1: Nautical Charts and Other Resources

1.1. Purpose and Role of Nautical Charts

1.1.1. Chart Categories

- *Sailing charts*

¹ From *U.S. Coast Pilot*®.

- General charts
- Coast charts
- Harbor charts

1.1.2. Types of Charts

2

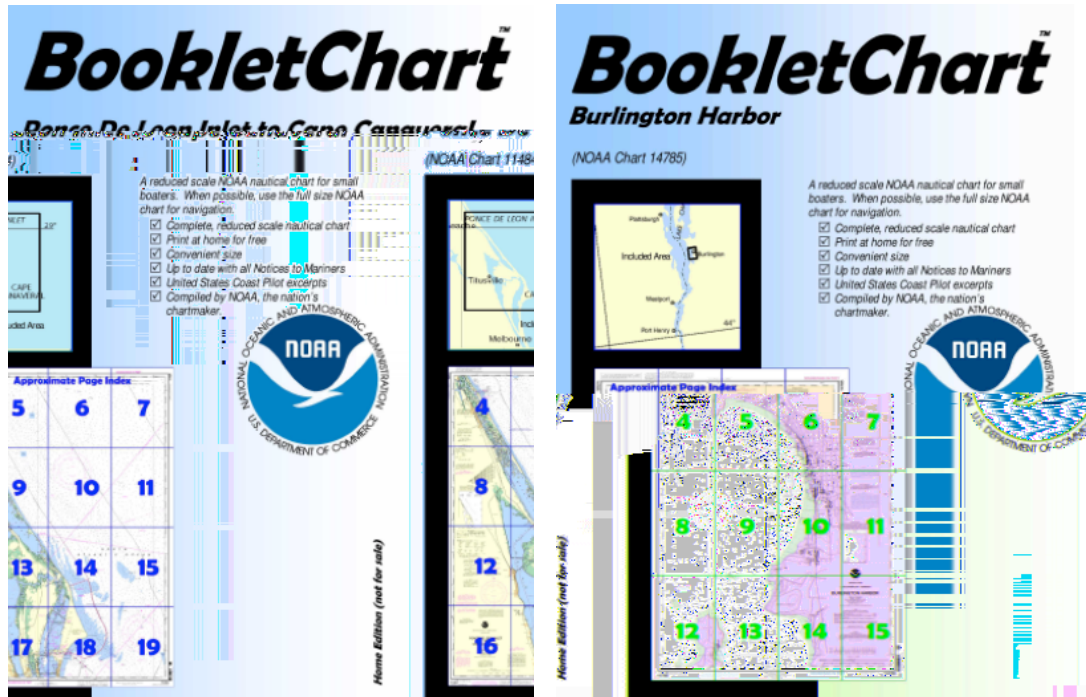


Figure 1.1. NOAA Booklet Charts 11484 (Ponce De Leon Inlet to Cape Canaveral) and 14785 (Burlington Harbor).

² <http://www.nauticalcharts.noaa.gov/>

print on demand (POD)

Booklet charts

3

4

5

6

-
-

³ Most of the chart fragment examples in this document come from booklet charts.

⁴ <https://iho.int/>

⁵ <https://navigation.usace.army.mil/Survey/InlandCharts>

⁶ <https://devgis.charttools.noaa.gov/pod>

-
-
-

RNC	ENC
Each pixel in the file is assigned a color and location; there is no other information	Every node knows where it is and what it is, including data about navigational aids, features, etc.
All information about the chart features is printed on the chart	Full data about a location or symbol can be obtained by a query
Display shows only traditional chart symbols	Display can show a variety of defined and proprietary symbols
Generally oriented "north up"	The chart has no inherent orientation but symbols and words always face up properly

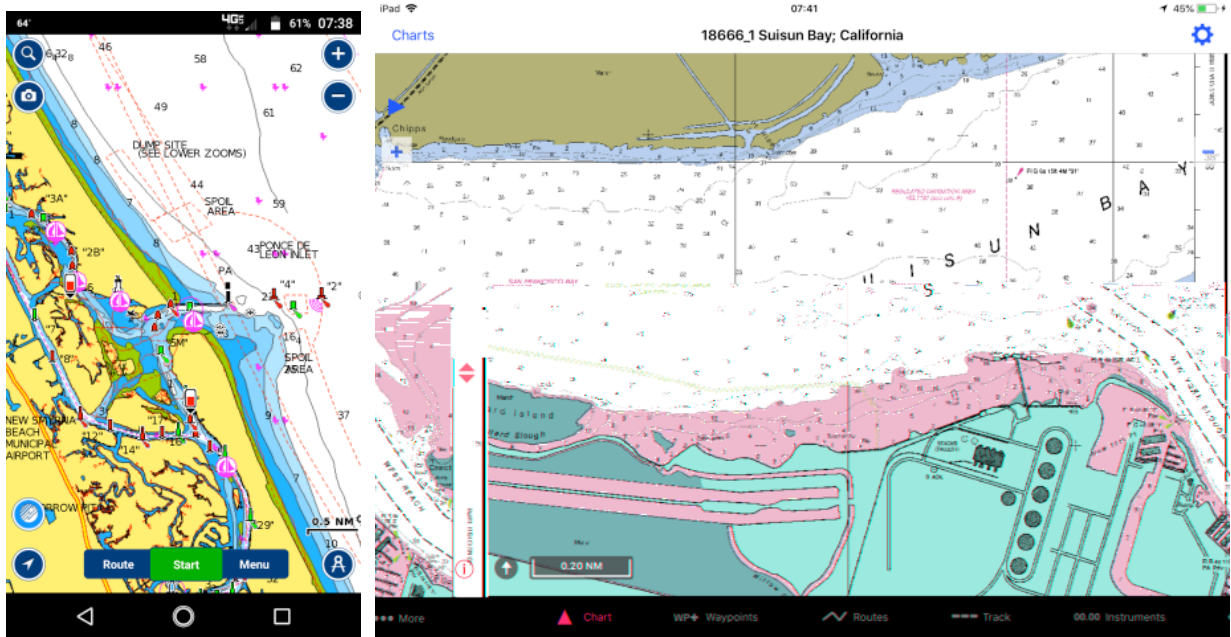


Figure 1.2. Screenshots of Navionics Boating HD (left, Android) and iNavX (right, iOS).

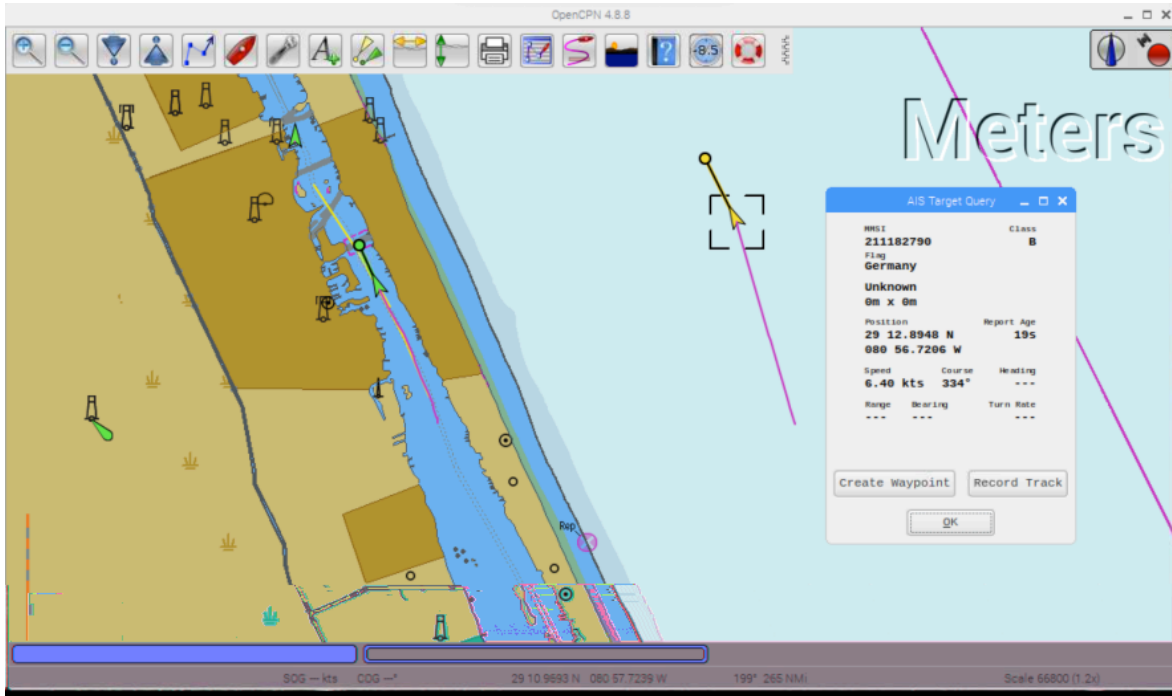


Figure 1.3. Screenshot of OpenCPN (Mac OS).

1.2. Supplementary Resources

1.2.1. Chart No. 1

Chart No. 1: Symbols, Abbreviations and Terms used on Paper and Electronic Navigational Charts

Chart No. 1

9

⁷ https://www.garykessler.net/library/ais_pi.html

⁸ <https://opencpn.org/>

⁹ <https://nauticalcharts.noaa.gov/publications/docs/us-chart-1/ChartNo1.pdf>

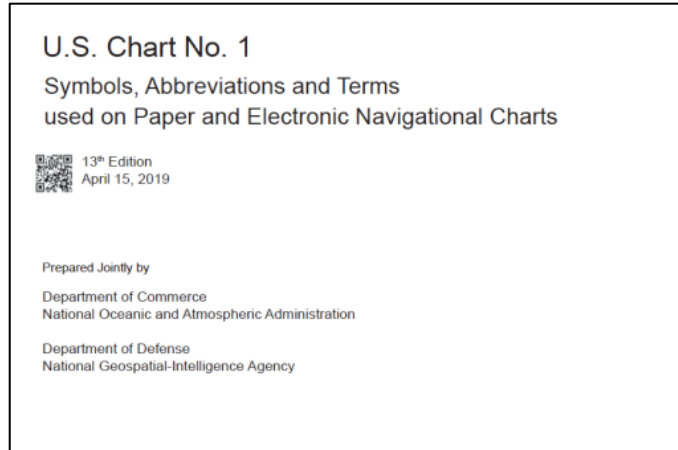


Figure 1.4. *Chart No. 1: Symbols, Abbreviations and Terms used on Paper and Electronic Navigational Charts.*

Chart No. 1

- *Introduction:*
- *General:*
- *Topography*
- *Hydrography:*
- *Navigation Aids and Services:*

1.2.2. U.S. Coast Pilot®

United States Coast Pilot®

U.S. Coast Pilot®

10

U.S. Coast Pilot®

Atlantic Coast Regions

¹⁰ <https://nauticalcharts.noaa.gov/publications/coast-pilot/index.html>

Great Lakes Region

Pacific Coast Regions



Figure 1.5. U.S. Coast Pilot® (Regions 4 and 6).

¹¹ There is also a section titled *Mineclearing: Caution*.

1.2.3. USCG Light List

U.S. Coast Guard Light List

Light List

Light List

12

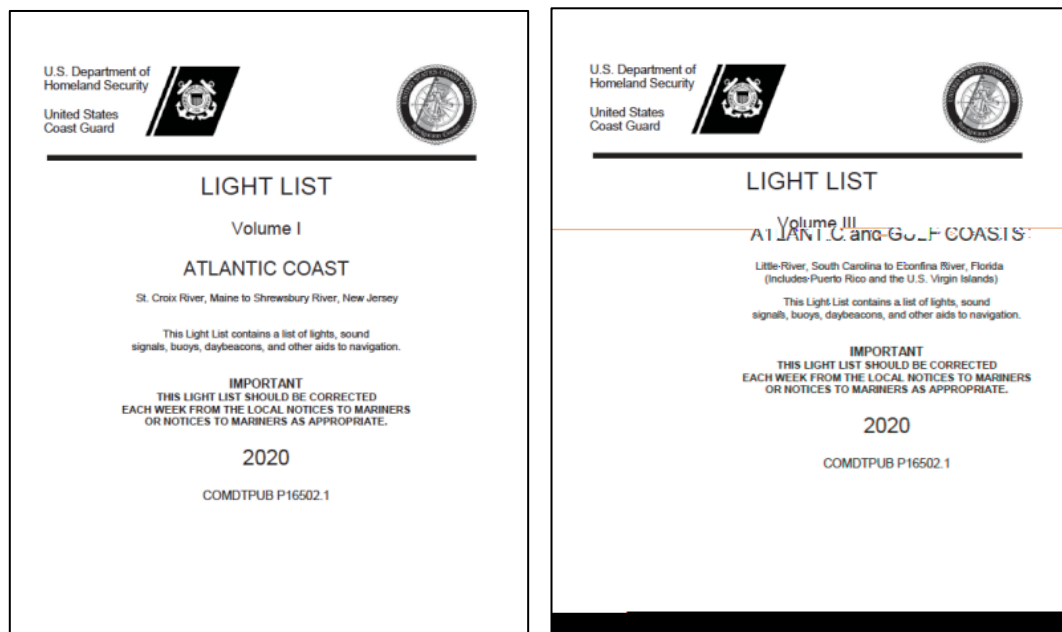


Figure 1.6. Light List (Volume I: Atlantic Coast and Volume III: Atlantic and Gulf Coasts).

¹² <http://www.navcen.uscg.gov/?pageName=lightlists>

1.2.4. Notice to Mariners

Coast Pilot® Light List

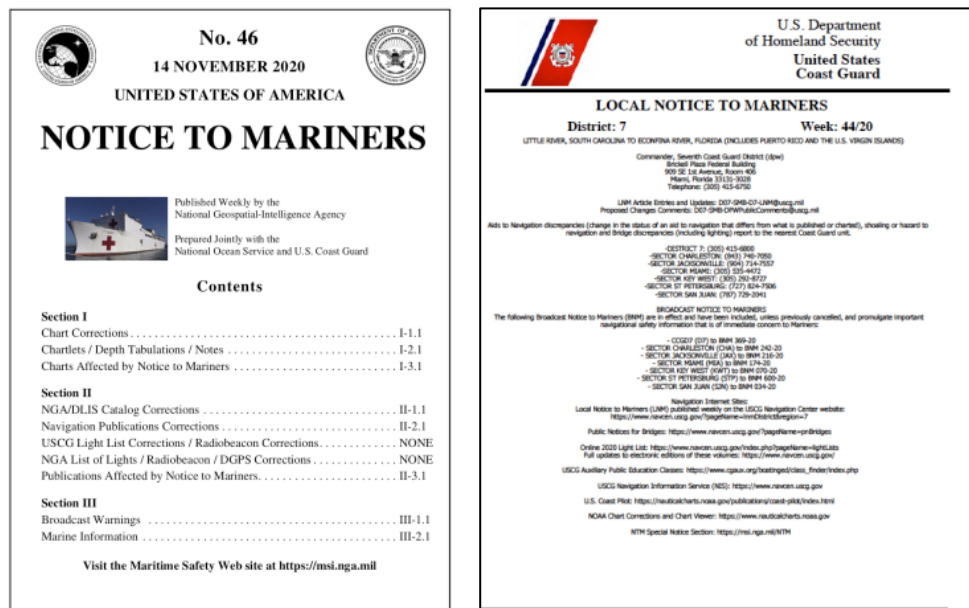


Figure 1.7. Notice to Mariners (No. 46 for the week ending 14 November 2020) and Local Notice to Mariners (44th week of 2020, first week of November, USCG District 7).

Notice to Mariners (NTM)

¹⁴ <http://www.navcen.uscg.gov/?pageName=lnmMain>

Part 2: Nautical Charts

2.1. Latitude and Longitude

latitude

parallel

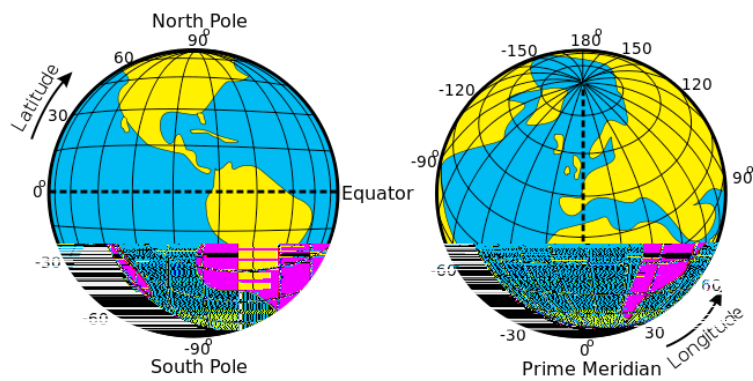


Figure 2.1. Latitude and longitude.¹⁵

longitude

meridian

¹⁵ http://en.wikipedia.org/wiki/File:Latitude_and_Longitude_of_the_Earth.svg

same. *Note that the latitude and longitude scales are not the*

minutes *seconds* *degree*
degree, minute, second
 $DD^{\circ}MM'SS''$ $DD^{\circ}MM.MM'$

$DD.DDD^{\circ}$

O.J. Walker

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2.2. Introduction to Aids To Navigation

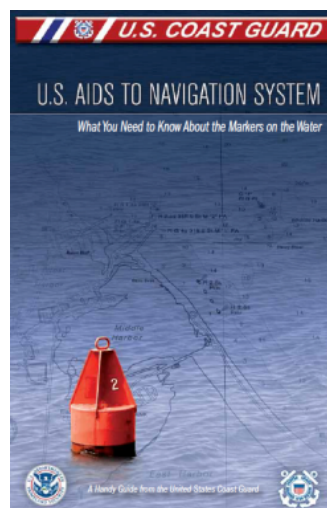


Figure 2.2. *U.S. Aids to Navigation* (USCG, 2011).

red right returning

U.S. Aids to

*Navigation System*¹⁸

¹⁶ <http://www.iala-aism.org/>

¹⁷ The IALA "A" system mnemonic is to match colors when returning from sea; i.e., green buoy on the right to match the boat's green light and red buoy on the left to match the boat's red light.

¹⁸ <http://www.uscgboating.org/images/486.PDF>

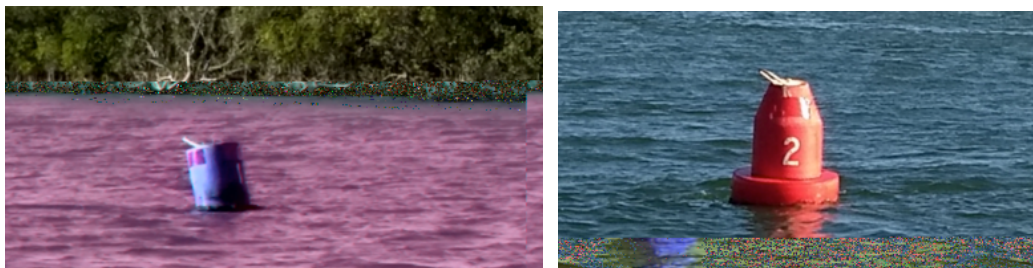


Figure 2.3. Green (can) and red (nun) buoys. [Photos by the author.]



Figure 2.4. Red triangular and green square dayboards. (Note the yellow square and light at the top of the post on #47. [Photos by the author.]

buoys

can

nun

beacons

*dayboard
light*

2.3. The Anatomy of a Nautical Chart

¹⁹ There are times when the ICW crosses a waterway marked according to the IALA lateral system. Although the day markers on the ICW are generally red with a yellow triangle and green with a yellow square, one might see a yellow triangle on a green can buoy or a yellow square on a red nun buoy when the ICW crosses a lateral system-marked channel. In those cases, the shape of the yellow symbol prevails as the ICW channel marker regardless of the color of the buoy (i.e., the yellow triangles mark the mainland side of the ICW even if on a green buoy).

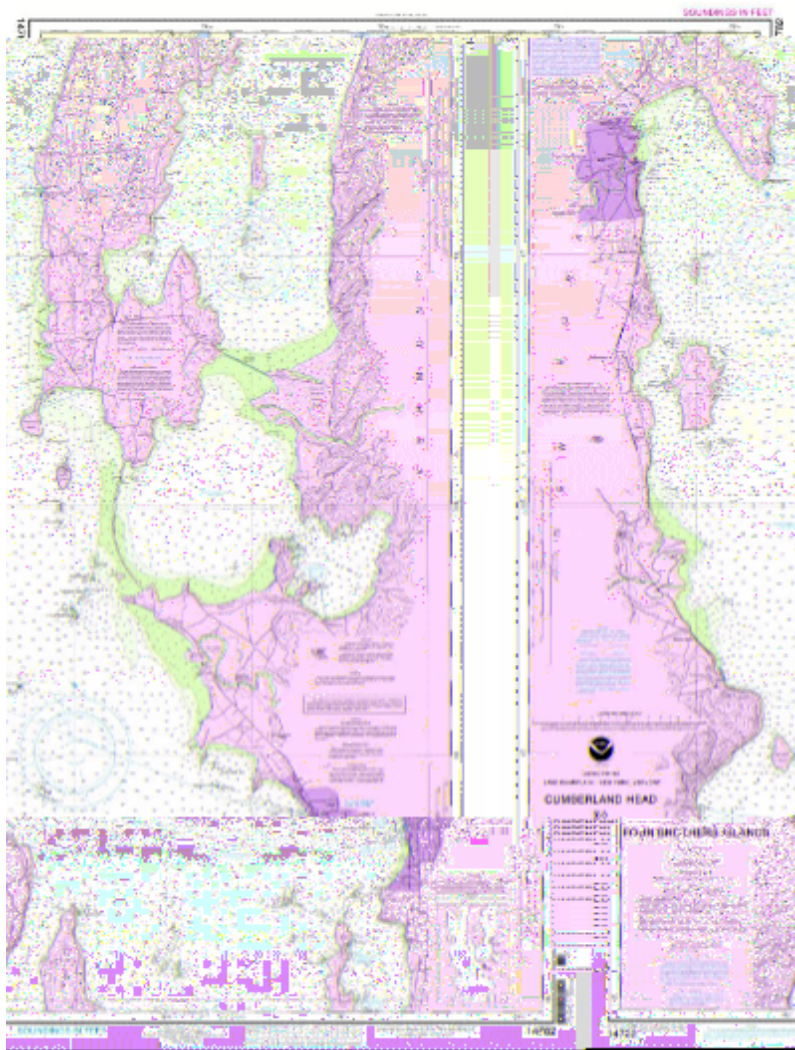


Figure 2.5. Chart 14782 (Cumberland Head to Four Brothers Islands).

2.3.1. Chart Reference Data

25th Ed., Jan. / 06 ■ Corrected through NM Jan. 14/06
Corrected through LNM Jan. 10/06
14782

CAUTION

This chart has been corrected from the Notice to Mariners (NM) published weekly by the National Geospatial-Intelligence Agency and the Local Notice to Mariners (LNM) issued periodically by each U.S. Coast Guard district to the dates shown in the lower left hand corner.

Figure 2.6. Chart number and date of publication (Chart 14782).

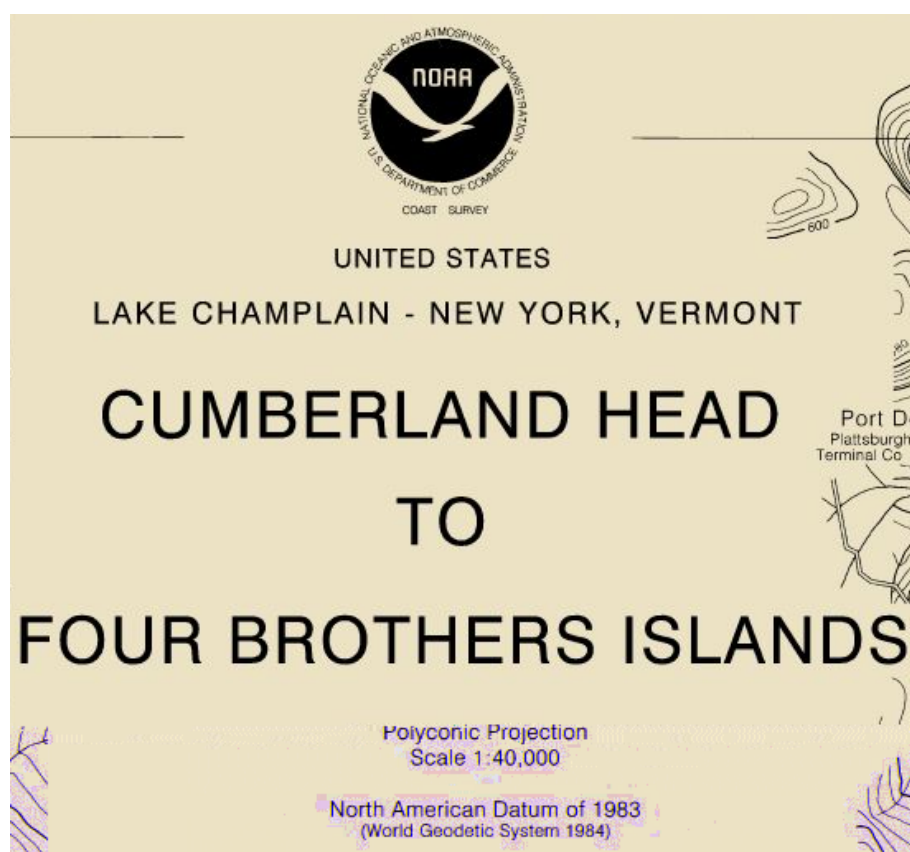


Figure 2.7. Chart title and other information (Chart 14782).

harbor chart



²⁰ The fact that one degree of longitude represents a different linear distance depending on latitude was a significant problem for mariners prior to the development of accurate clocks in the 1700s. For an excellent description of the Longitude Act of 1714 and the development of precise nautical clocks, see *Longitude* (Sobel, 1995).

2.3.2. The Compass Rose

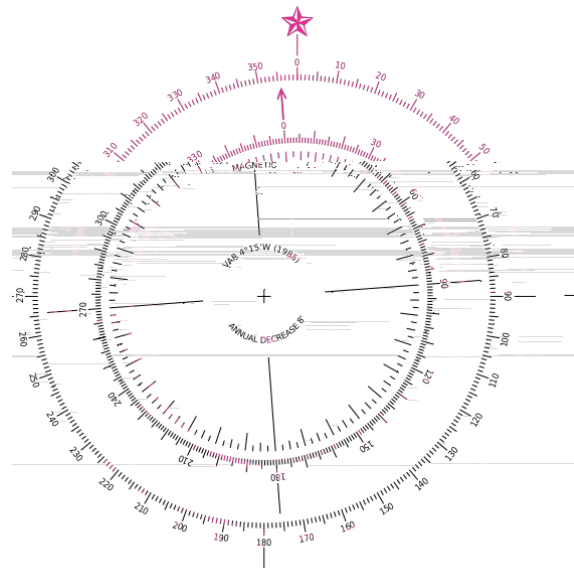


Figure 2.11. Compass rose.²²

²¹ Tide tables, which are not discussed in this tutorial, assist the mariner in knowing how the water depth fluctuates on a daily basis, which can aid in planning. The NOAA Tides & Currents Web page can be found at <http://tidesandcurrents.noaa.gov/>.

²² http://en.wikipedia.org/wiki/File:Modern_nautical_compass_rose.svg

(76) Burlington Breakwater North Light (44°28'50"N., 73°13'47"W.), 35 feet above the water, is shown from a white square lighthouse on the north end of the north breakwater.

2.3.4. *Marginal Notes*

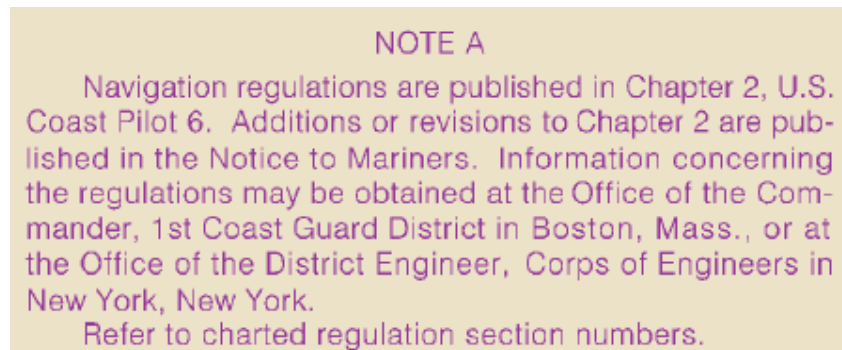


Figure 2.13. NOTE A from Chart 14782.

2.3.5. *Symbols for Aids to Navigation*

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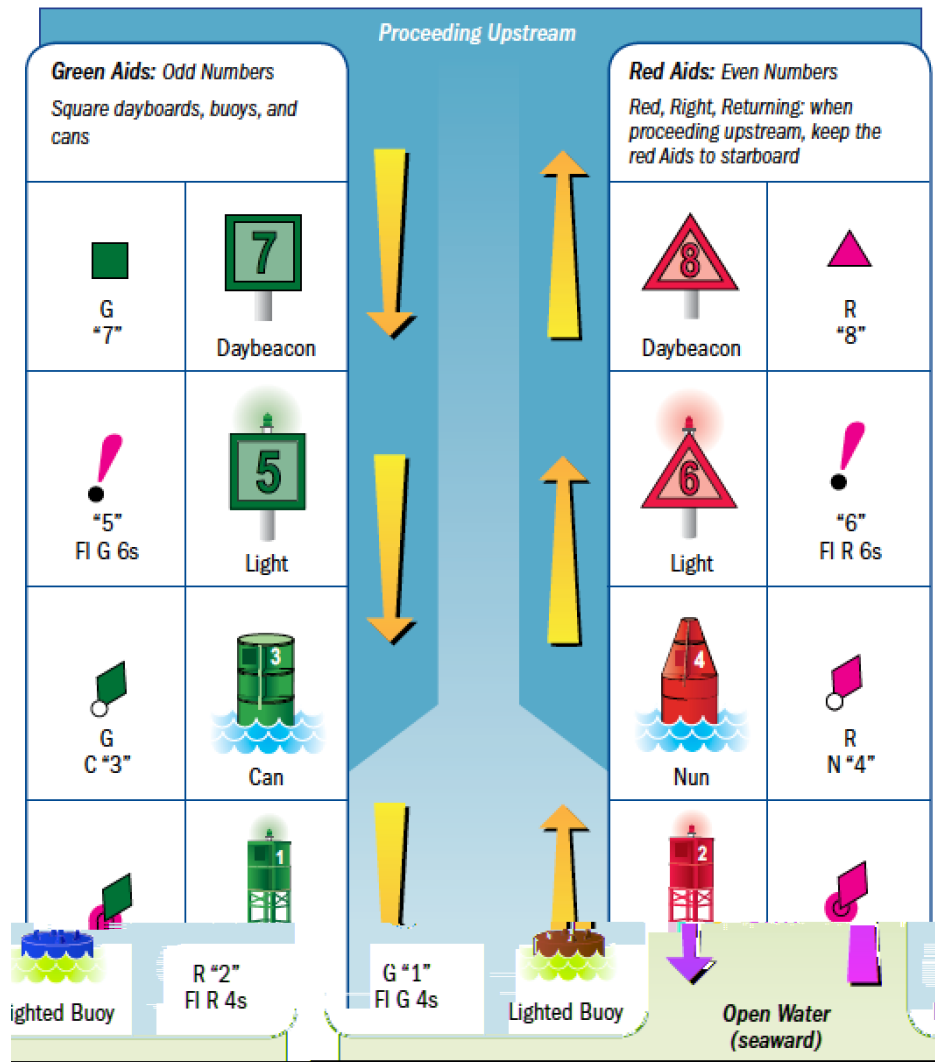


Figure 2.14. Channel markers and symbols.
(From U.S. Aids to Navigation System).

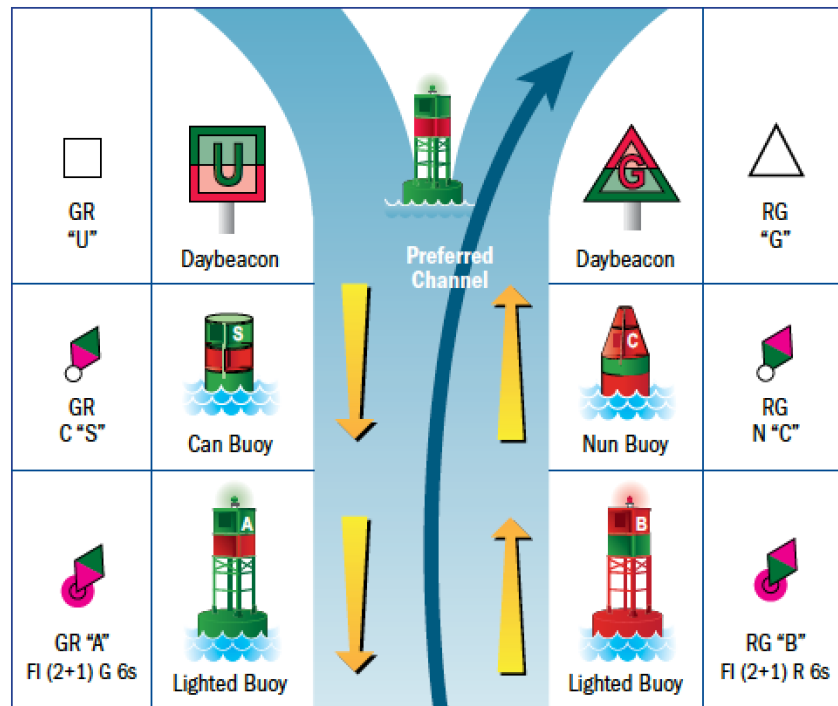


Figure 2.15. Preferred channel markers and symbols.
 (From *U.S. Aids to Navigation System*).

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2.3.6. Examples from the Charts

Chart No. 1

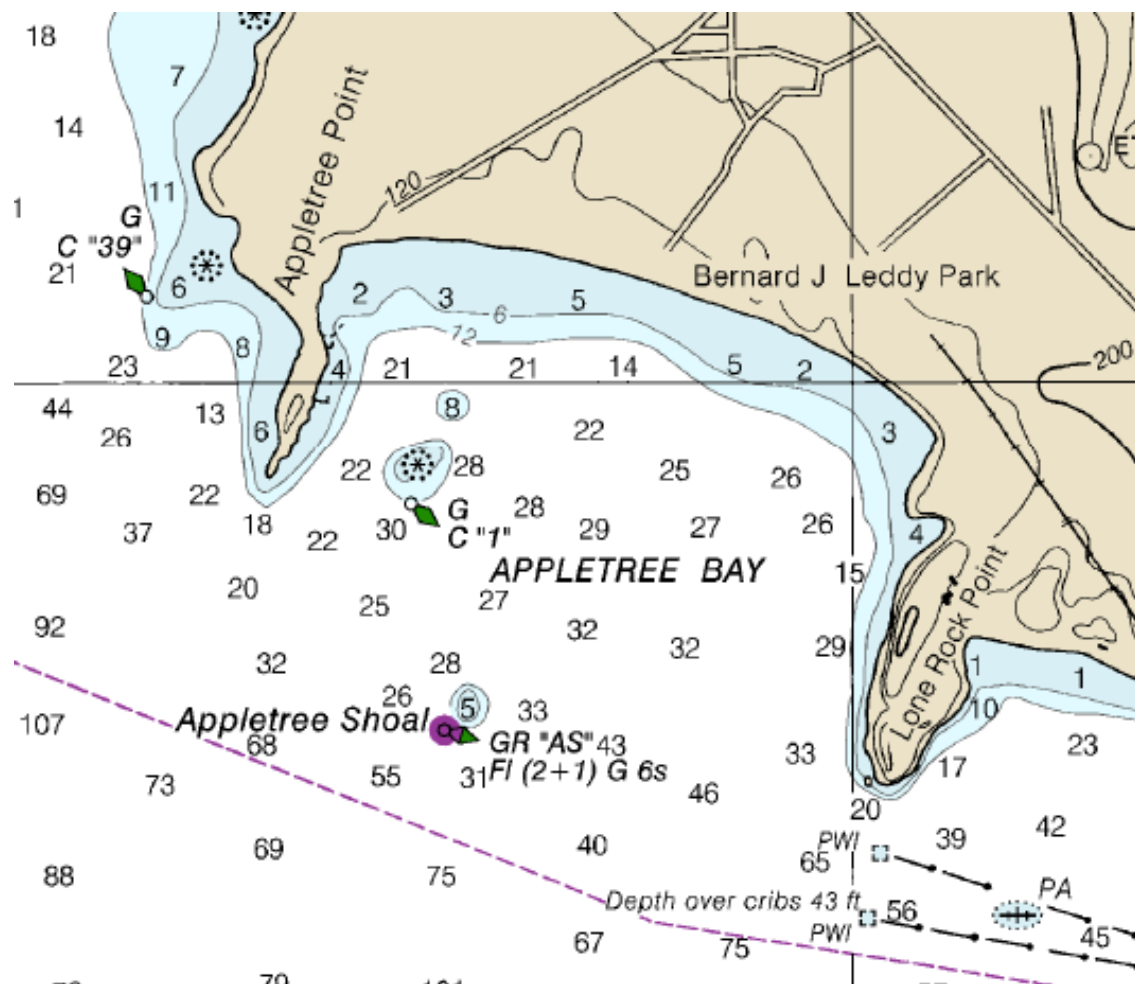


Figure 2.16. Appletree Bay, from Chart 14782.

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24

25

Chart No. 1

Chart No. 1

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26

27

²⁴ The wreck indicated is called the *Horse Ferry*, the only known wreck of a horse-powered vessel in North America.

²⁵ This ferry runs between Burlington and Port Kent, New York. Knowing the route of a scheduled ferry is important as it has right-of-way over most other boats in the area.

²⁶ Lake Champlain information is in *Volume I: Atlantic Coast (St. Croix River, Maine to Shrewsbury River, New Jersey)*.

²⁷ Note that latitude and longitude are expressed to the thousandth of a second, which is roughly 1.2 inches (3 cm).

39605	Appletree Point Buoy 1 At southeast side of 8-foot spot.	44-29-47.174N 073-16-05.963W
39610	Appletree Shoal Lighted Buoy AS	44-29-23.175N 073-15-59.963W

FI (2+1)G 6s

Green and red
bands.

[illegible]

●

●

●

●

●

²⁸ This wreck is called the *Phoenix*.

39540	COLCHESTER REEF LIGHT	44-33-18.400N 073-19-44.300W	Fl W 4s	51	6	NG on post on granite caisson.	
39565	Colchester Shoal Lighted Buoy	44-32-58.070N 073-19-06.670W	Fl G 2.5s		3	Green.	Maintained from May 1 to Nov. 1.
lov. 39570	Colchester Shoal Buoy 37 At southeast end of shoal.	44-32-47.920N 073-19-57.713W				Green can.	Maintained from May 1 to N 1.

Lake Champlain Underwater Preserve

Aids maintained from June 1 to Nov. 1. Aids mark historic shipwreck diving sights.

39655	- Buoy E	44-33-01.000N 073-19-48.000W				Yellow sphere.	Private aid.
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Fl R 2.5s 35 ft 14 St M

W Bn

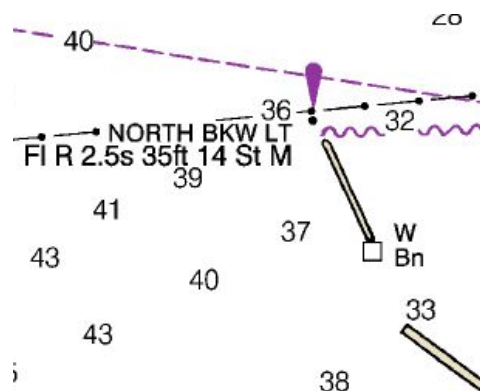


Figure 2.18. Burlington harbor breakwater, from Chart 14785 (Burlington Harbor).

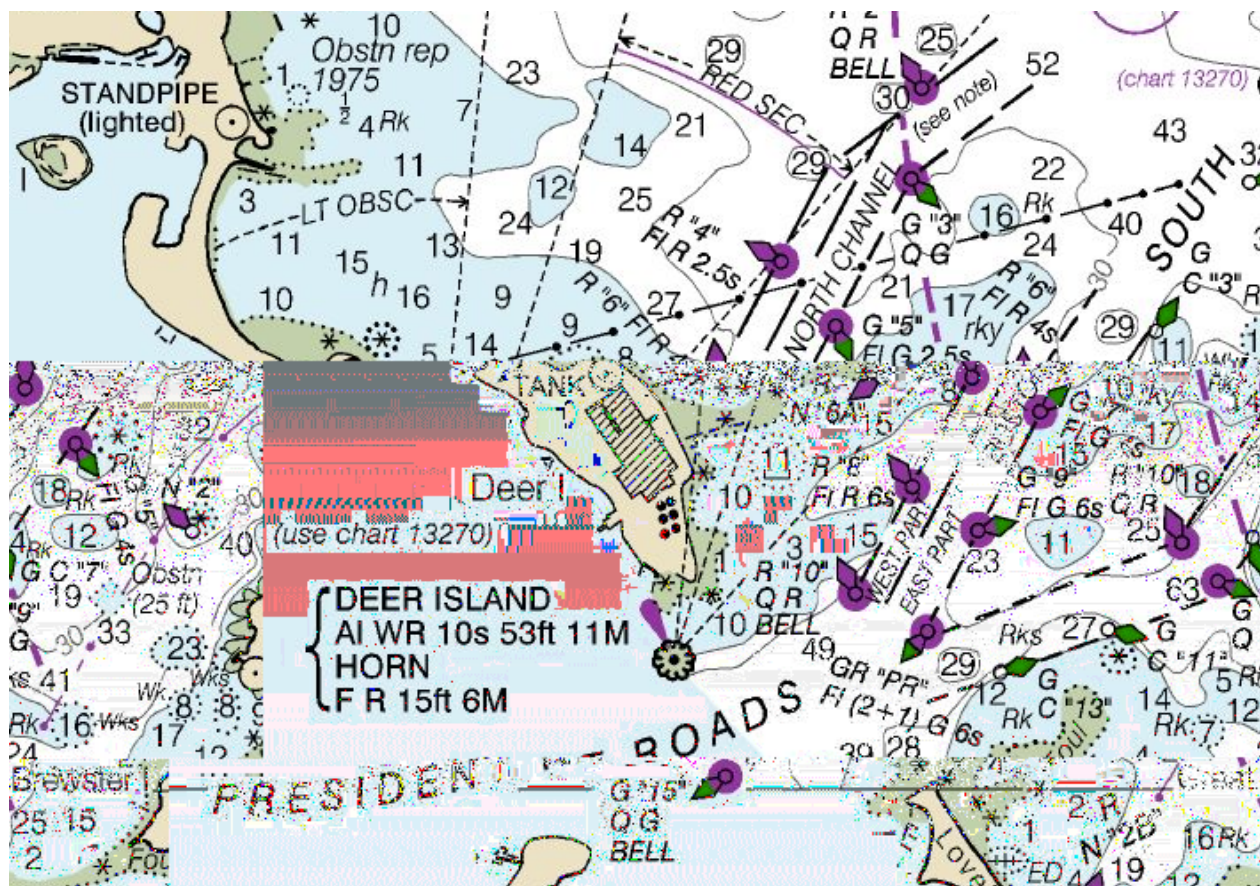


Figure 2.19. Deer Island and President Roads section of Chart 13267 (Massachusetts Bay).

Light List

10795	Deer Island Light On south end of spit.	42-20-23.461N 070-57-16.225W	AI WR 10s	53	11	Red cylindrical tower; black cylindrical pier. 53	Obscured from 112° to 186°. Red danger sector from 198° to 222°. HORN: 1 blast ev 10s (1s bl).
10797	DEER ISLAND DANGER LIGHT On south end of spit.	42-20-23.400N 070-57-16.000W	F R	15	6		Red from 198° to 222°.

from the perspective of the vessel

Q G

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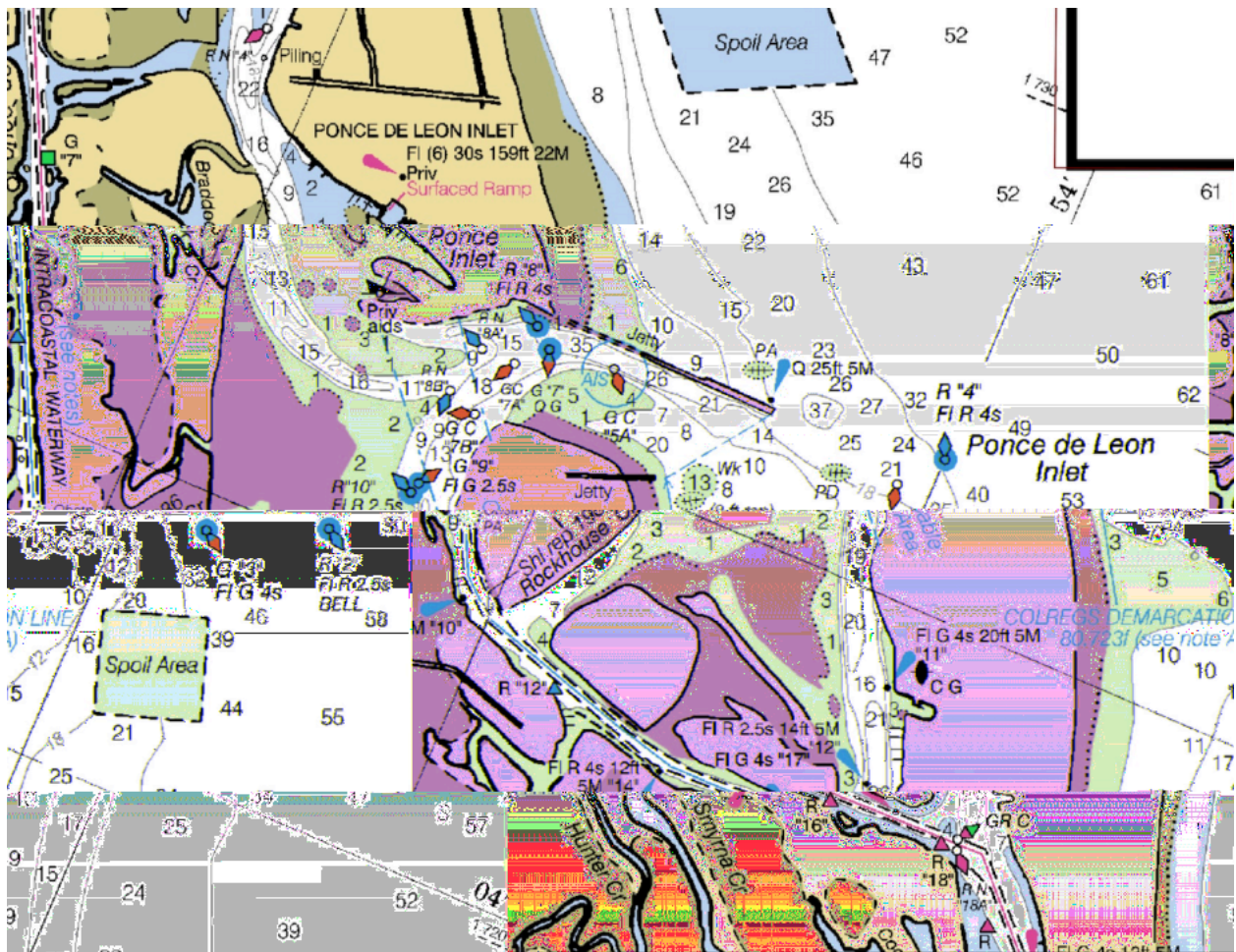


Figure 2.20. Area around Ponce de Leon Inlet from Chart 11485
(Intracoastal Waterway Tolomato River to Palm Shores).

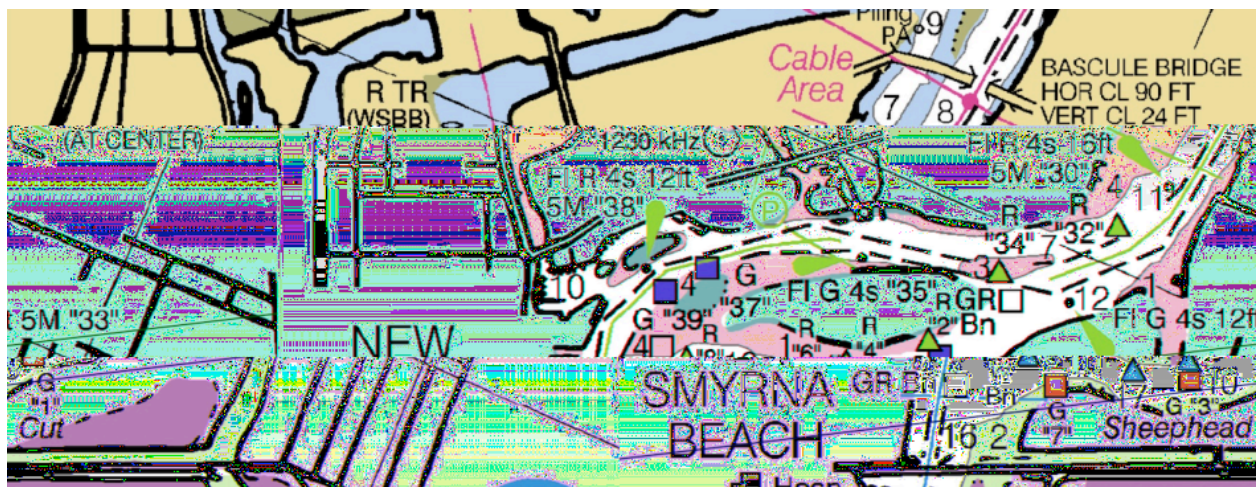


Figure 2.21. Area around Chicken Island in New Smyrna Beach from Chart 11485.

basculer bridge

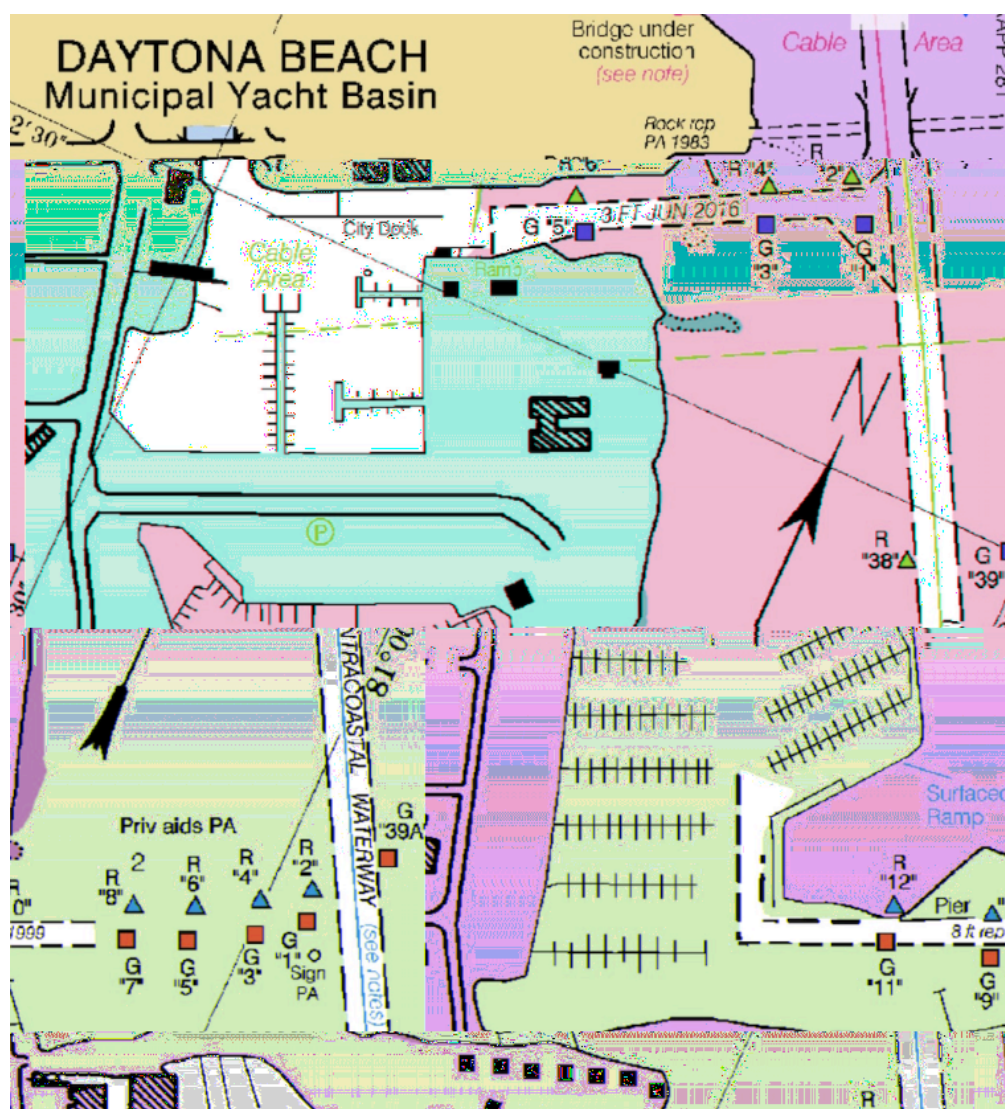


Figure 2.22. Daytona Beach Municipal Yacht Basin inset from Chart 11485.

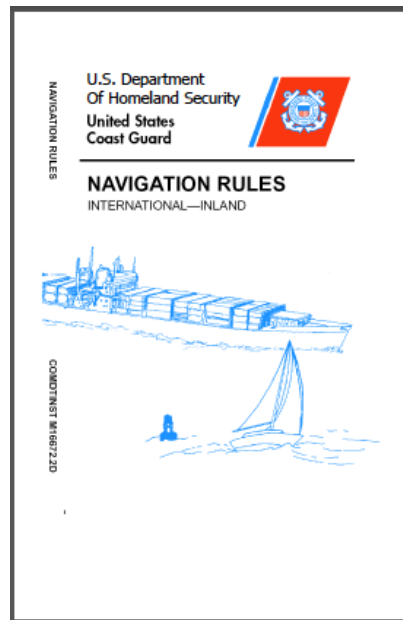


Figure 2.23. *Navigation Rules and Regulations Handbook* (USCG, 2014).

*Regulations Handbook*³⁰

*Navigation Rules and
colregs*

²⁹ This chart shows a bridge under construction. The former Veterans Memorial Bridge was a bascule bridge, demolished during the summer of 2016, and still shows on some charts. A new, fixed bridge opened in August 2020. Be sure to always use the most up-to-date information!

³⁰ http://www.navcen.uscg.gov/pdf/navRules/Handbook/CG_NAV_RULES_20140910.pdf

2.4. Summary

How to Read a Nautical Chart

Part 3: Plotting

3.1. Which Way is North?

north

3.1.1. The Compass Rose, Revisited

not

variation

wandering magnetic pole ³¹

³¹ See NOAA (n.d., 2019) for more information about the wandering magnetic north pole. Historically, the shift has been approximately 10 miles (16 km) per year; current models show a change of 34 miles (55 km) per year.

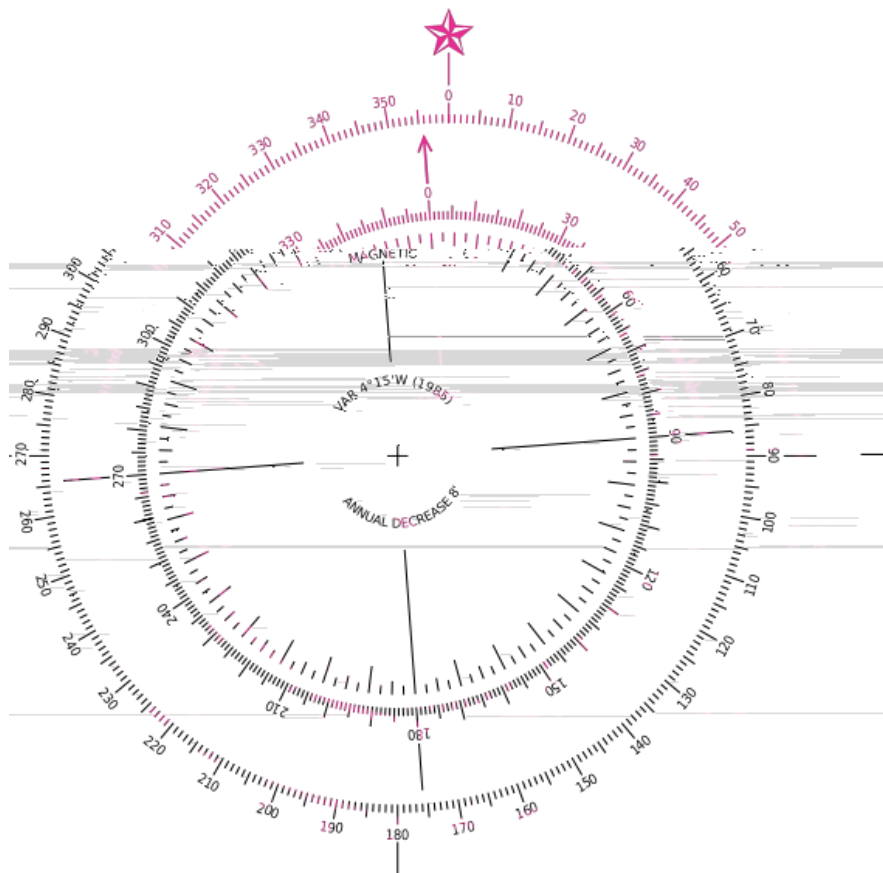


Figure 3.1. Compass rose.³²

3.1.2. Deviation

deviation

³² http://en.wikipedia.org/wiki/File:Modern_nautical_compass_rose.svg

Heading	Deviation	Heading	Deviation	Heading	Deviation
0°	1°E	120°	1°E	240°	2.5°W
30°	1.5°E	150°	0°	270°	2°W
60°	2°E	180°	1°W	300°	0°
90°	3°E	210°	2°W	330°	0.5°E

3.1.3. Putting it all Together

course per steering compass

uncorrecting the compass

correcting the compass

33

3.2. Instruments for Plotting

3.2.1. Parallel Rule and Roller Plotter

parallel ruler

34



Figure 3.2. Parallel ruler.

³³ One such site is <http://captnmike.com/2011/11/13/tvmc-%E2%80%93-practice-%E2%80%93-correcting-and-un-correcting-the-compass/>.

³⁴ Instruments shown here are from Weems & Plath. Pictures are for example purposes only and should not be interpreted as a recommendation or product endorsement.

<http://www.boatsafe.com/navigation/rules1.htm>

<http://www.youtube.com/watch?v=xKSBYfscA80>

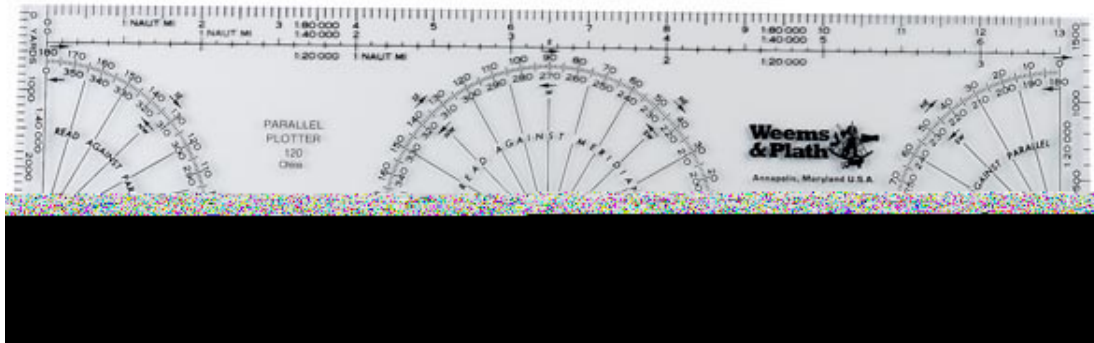


Figure 3.3. Parallel, or roll, plotter.

parallel, or roll, plotter



Figure 3.4. Protractor triangle.

protractor triangle

3.2.2. Dividers and Compass

dividers



Figure 3.5. Dividers.

compass

3.3. Arithmetic for Plotting

3.3.1. Measuring Distance

latitude

Never use the longitude (horizontal) scale to determine the distance between two points.

Figure 3.6. Placing the dividers on two points.

Figure 3.7. Placing the dividers on the latitude scale.

<http://www.boatsafe.com/navigation/divide1.htm>

3.3.2. Manipulating Time Values

-
-
-
-

$$\rightarrow \begin{array}{c} -1 + 60 \\ \hline \end{array} \rightarrow \begin{array}{c} 19 \ 70 \\ \hline \hline \end{array}$$

$$\rightarrow \begin{array}{c} +24 \\ \hline \end{array} \rightarrow \begin{array}{c} 25 \\ \hline \end{array}$$

-
-
-

- ×
- ×
- ×

-
-
-

3.3.3. Distance, Speed, and Time

-
-
-

35

36

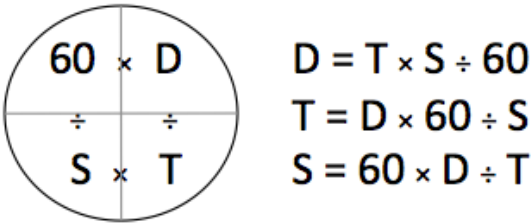


Figure 3.8. 60 D ST mnemonic.

60 D ST

³⁵ Note that the speedometer on many small recreational boats uses miles per hour (MPH) as the unit of measurement rather than knots. If you perform these calculations using MPH, distances will be in statute miles rather than nautical miles.

³⁶ The abbreviation *kt* is also commonly used for knots although *kn* is the international standard.

Example 1:

×

Example 2:

×

Example 3:

×

nautical slide rule

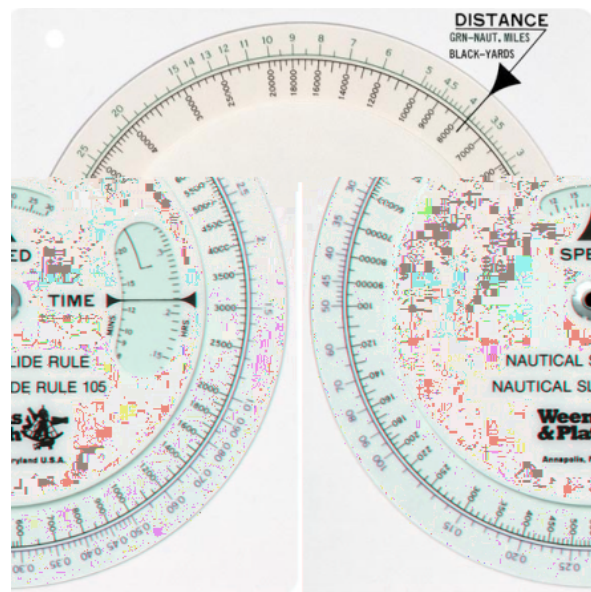


Figure 3.9. Nautical slide rule.

3.4. Plotting Problems

For the examples in this section, make the assumption that it is the year 2002. According to the compass rose on the chart, the variation was 4°15'W in 1985 with an annual decrease of 8'; after 17 years, the variation would have decreased by 2°16' to be approximately 2°W.

Similarly, the following deviation table will be used for all examples:

Heading	Deviation	Heading	Deviation	Heading	Deviation
0°	1°E	120°	1°E	240°	2.5°W
30°	1.5°E	150°	0°	270°	2°W
60°	2°E	180°	1°W	300°	0°
90°	3°E	210°	2°W	330°	0.5°E

The chart in Appendix E is wholly fictitious. Do not attempt to relate this chart to anything in the real world.

³⁷ E.g., suppose you know the distance to your destination, course heading, and speed, and are asked to determine how long the trip will take. To solve for time, you need only know distance and speed; heading is irrelevant.

³⁸ From <http://captnmike.com/2009/10/01/piloting-and-navigators-quick-reference/>

3.4.1. Finding Latitude and Longitude

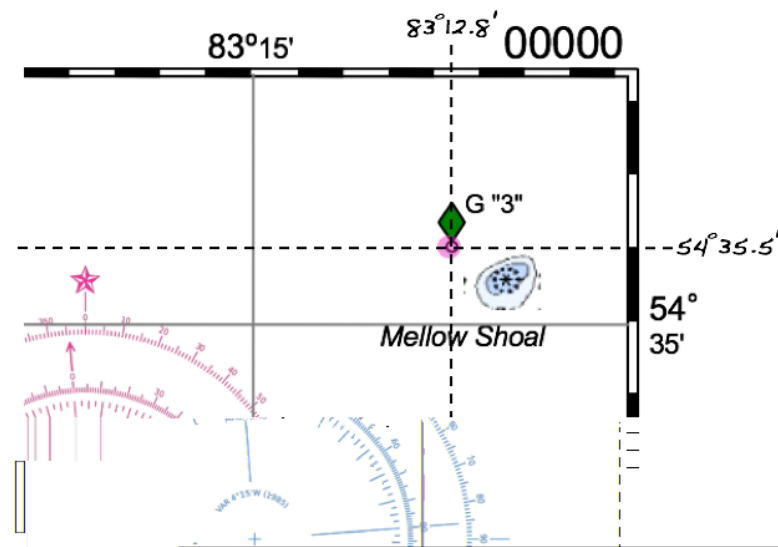


Figure 3.10. Finding latitude and longitude.

[http://www.youtube.com/watch](http://www.youtube.com/watch?v=K02gFTAz5Mo)

?v=K02gFTAz5Mo

As an aside, note that this chart does not specifically indicate whether the location is north or south of the Equator, or east or west of the Prime Meridian. You can determine this information, however, from the context of the latitude and longitude markings on the chart. Remember that

charts are always oriented with true north towards the top. When looking at this chart, notice that the minutes on the vertical scale increase going from bottom to top; this means that this chart is in the north latitudes. Similarly, the minutes along the horizontal scale are increasing from right to left, meaning that the chart is in the western longitudes.

3.4.2. Plotting a Position

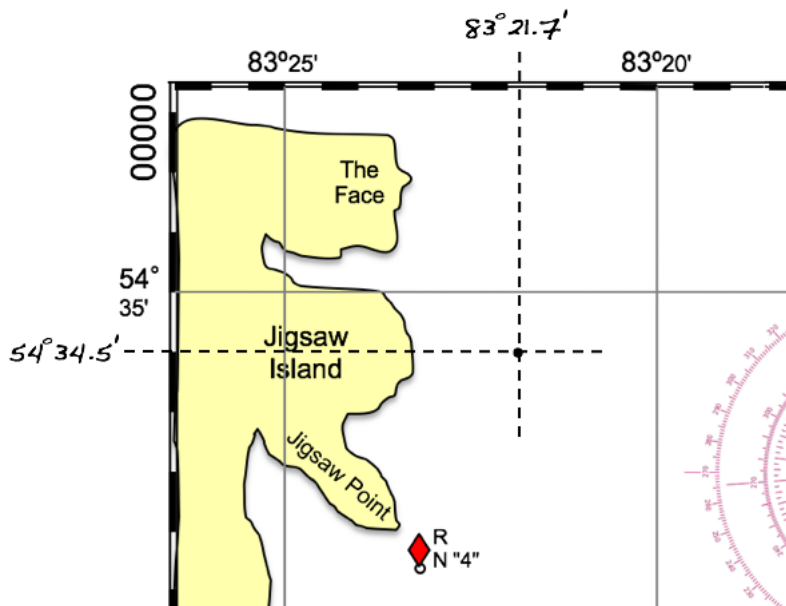


Figure 3.11. Plotting position.

3.4.3. Finding a True Bearing and Compass Course³⁹

all

³⁹ This section describes how to find the compass course knowing an initial point and an intended destination; another term for this is the *intended course*. If you know the two points and want to find the course between them, this is sometimes referred to as the *course made good*, which is the actual course you took between the two points after accounting for compass variance and deviation, wind, tide, current, and other factors. Bowditch (2002) recommends using the phrase *track made good*.

trackline

outer

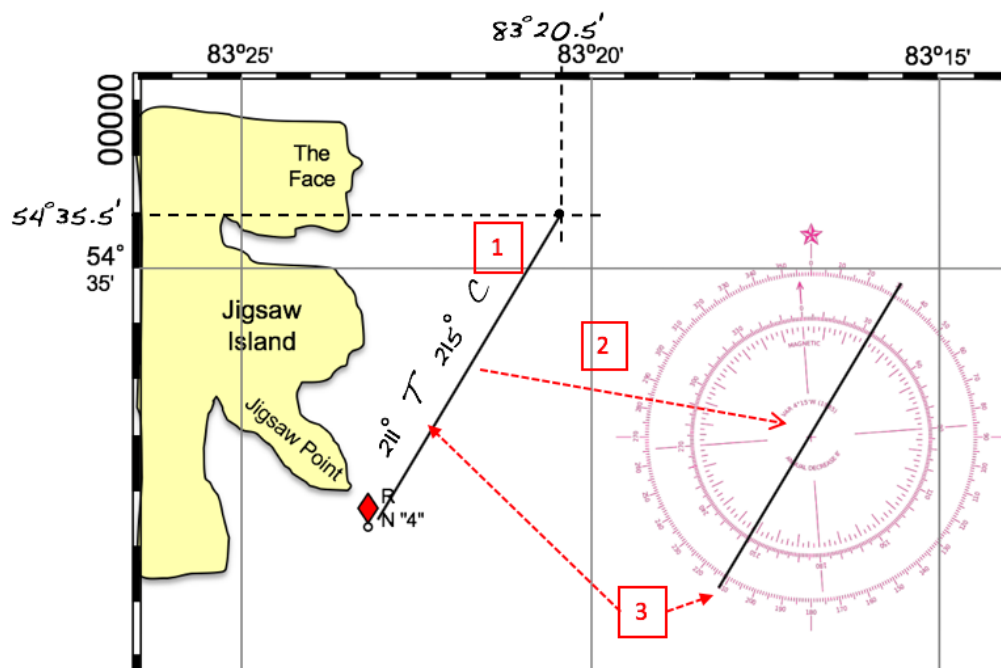


Figure 3.12. Finding a true bearing and compass course.

compass course

TVMDC + W

3.4.4. Finding Distance and Speed Made Good

vertical (latitude)

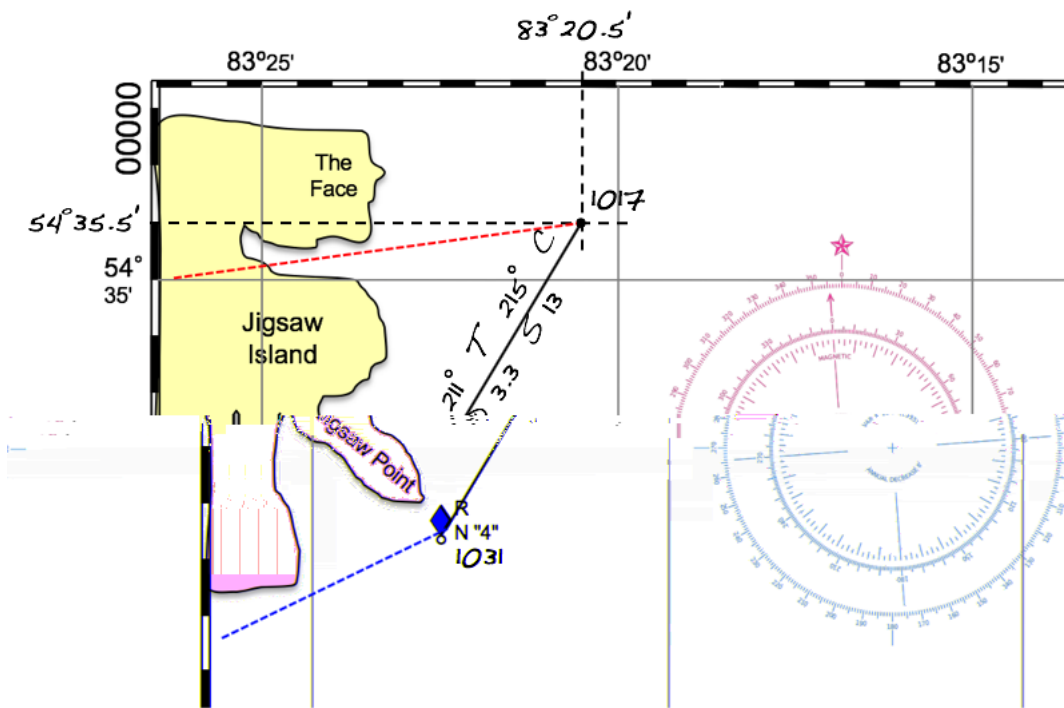


Figure 3.13. Finding distance and speed made good.

60 D ST

×

×

3.4.5. Estimating Time of Arrival

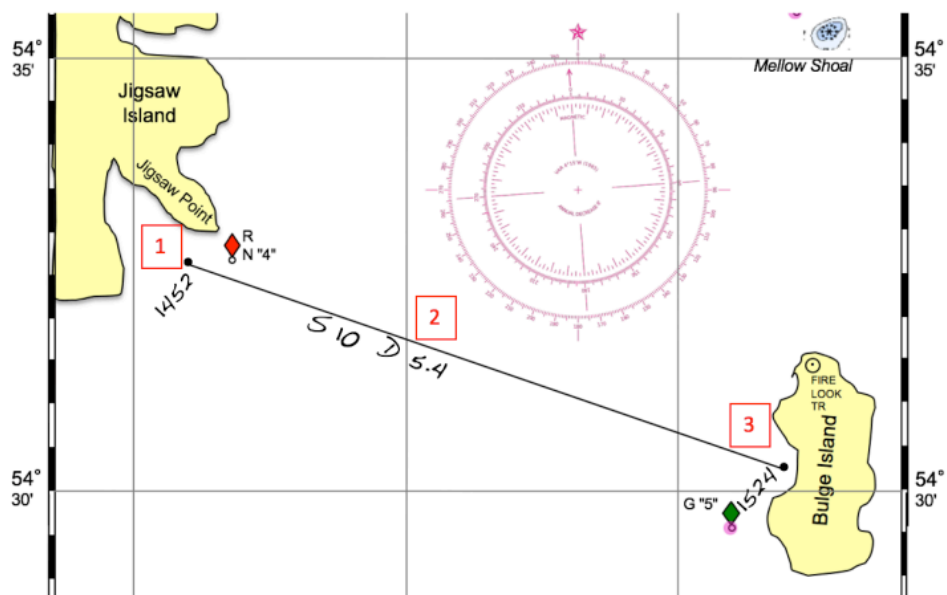


Figure 3.14. Estimating time of arrival.

3.4.6. Dead Reckoning

60 D ST

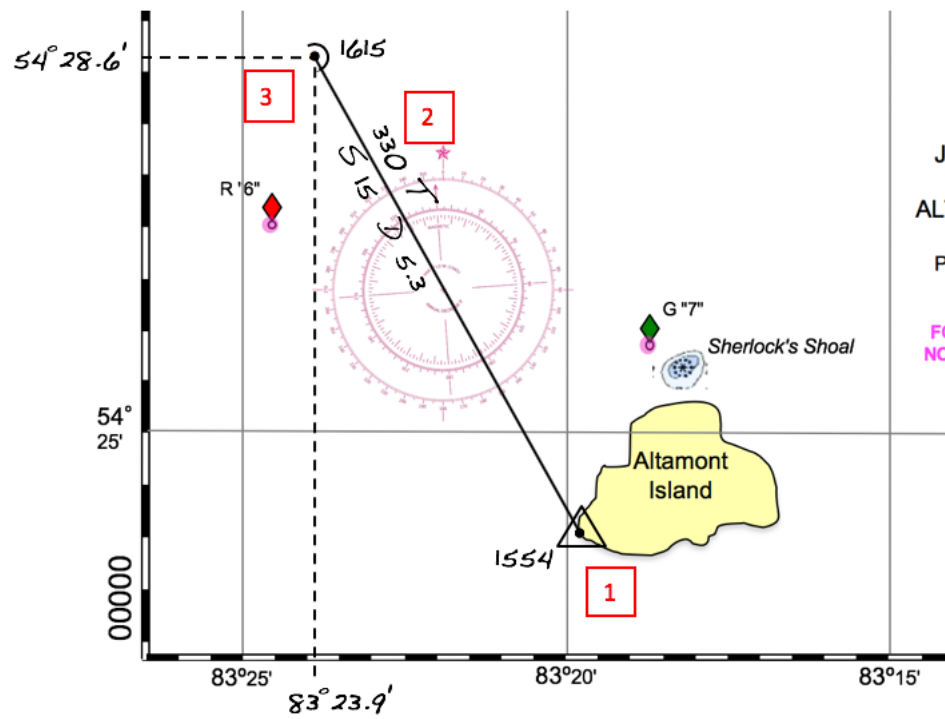


Figure 3.15. Dead reckoning.

3.4.7. Finding Set and Drift

set and drift

drift

40

Set

60 D ST

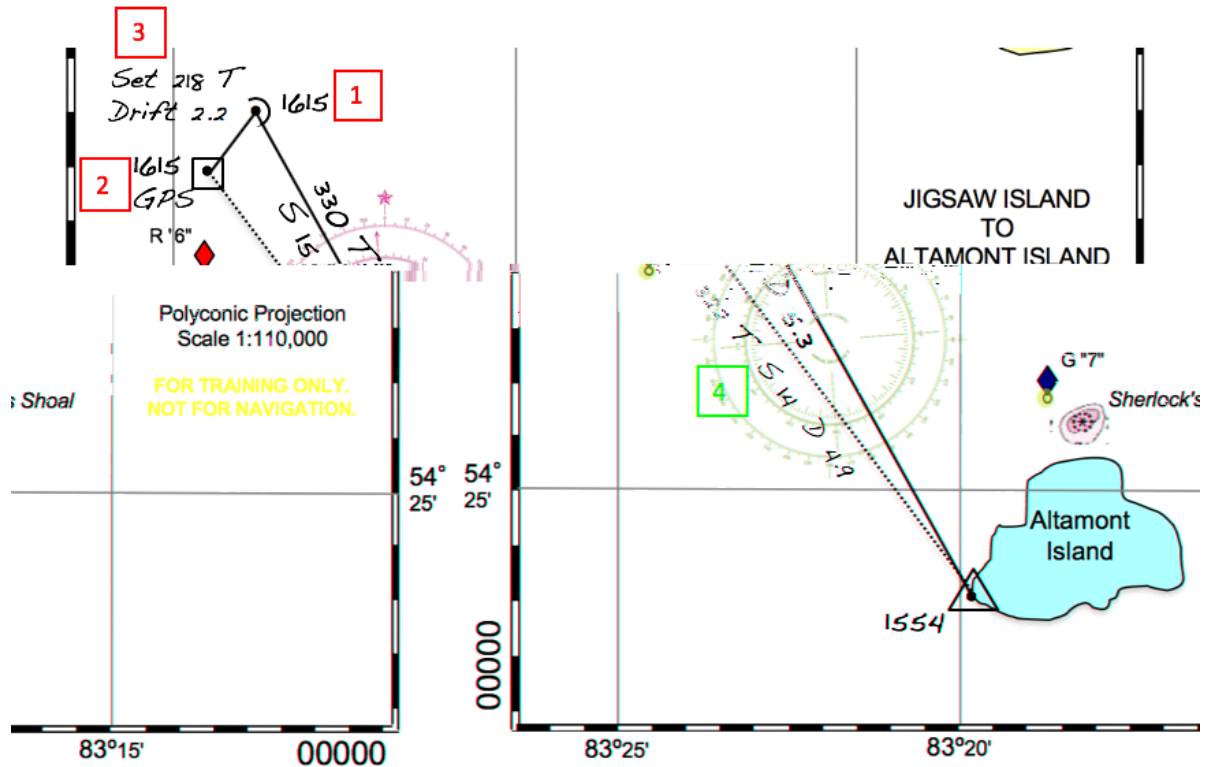


Figure 3.16. Determining set and drift.

⁴⁰ Without getting too much into the topic of vector analysis, some readers might have already recognized that a trackline is a vector. In physics, a vector is a line that has both direction (e.g., heading) and magnitude (e.g., distance). Adding the two vectors of a dead reckoning trackline and "set and drift" yields the actual position.

3.4.8. Plotting A Course With Known Set and Drift

point to aim (PTA)

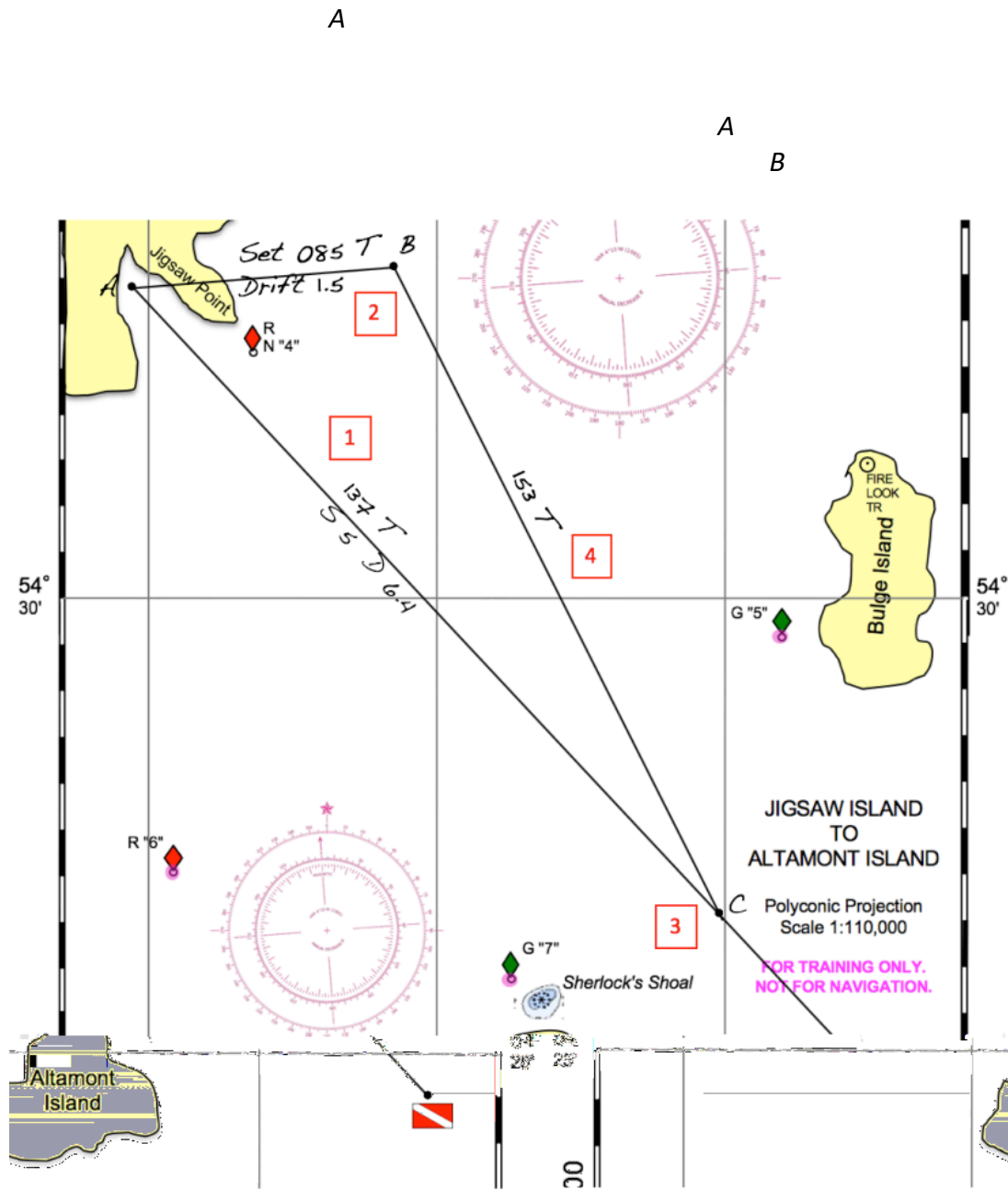


Figure 3.17. Plotting with known set and drift.

C

B C

TVMDC + W

3.4.9. Calculating ETA With Known Set and Drift

WARNING: Some new arithmetic ahead!

41

×

⁴¹ You can confirm this using the 60 D ST formula where D = 4.2 nm and S = 5 kn.

3.4.10. Obtaining a Fix With Lines of Position

line of position (LOP)

In the example in this section, the LOP headings are obtained using a hand-held compass. To convert to true north, we have to correct the magnetic heading by the variation. We can ignore the effects of the vessel itself, however, so deviation will not be an issue.

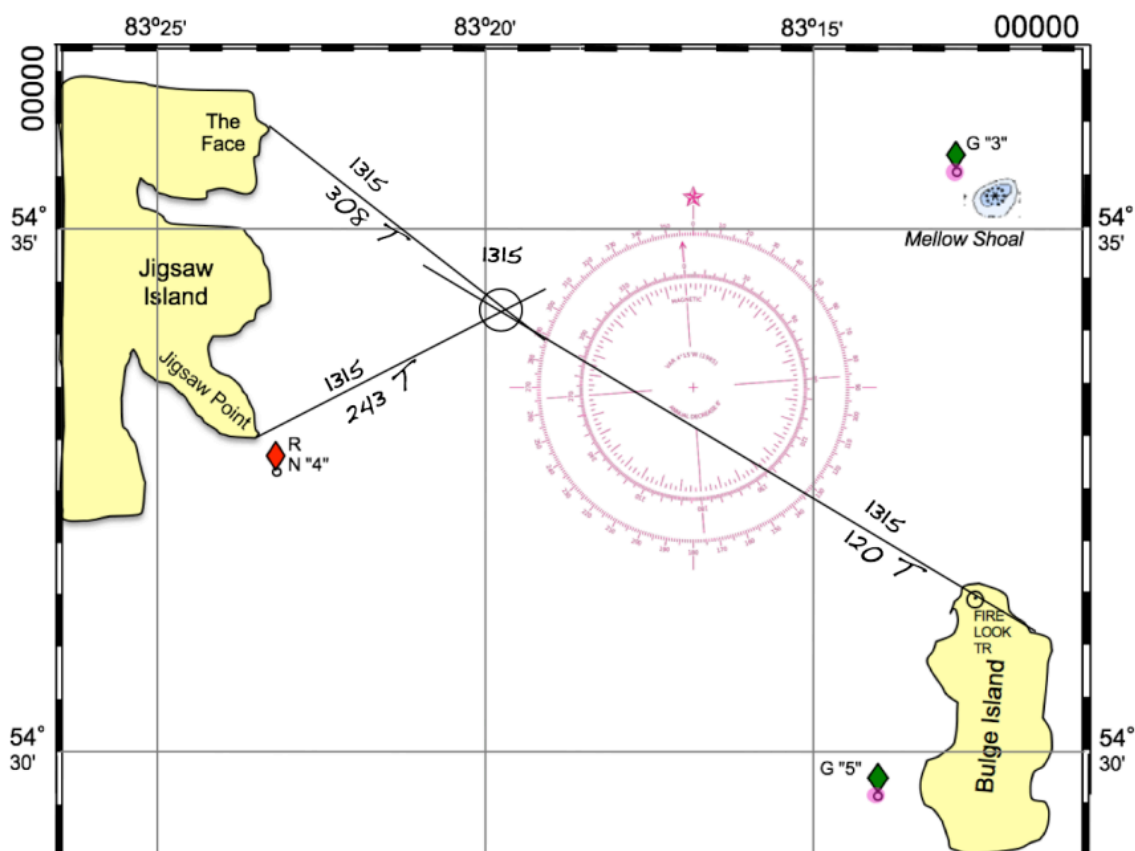


Figure 3.18. Obtaining a three-bearing fix.

LOP Target	Compass	True

3.4.11. Taking a Running Fix

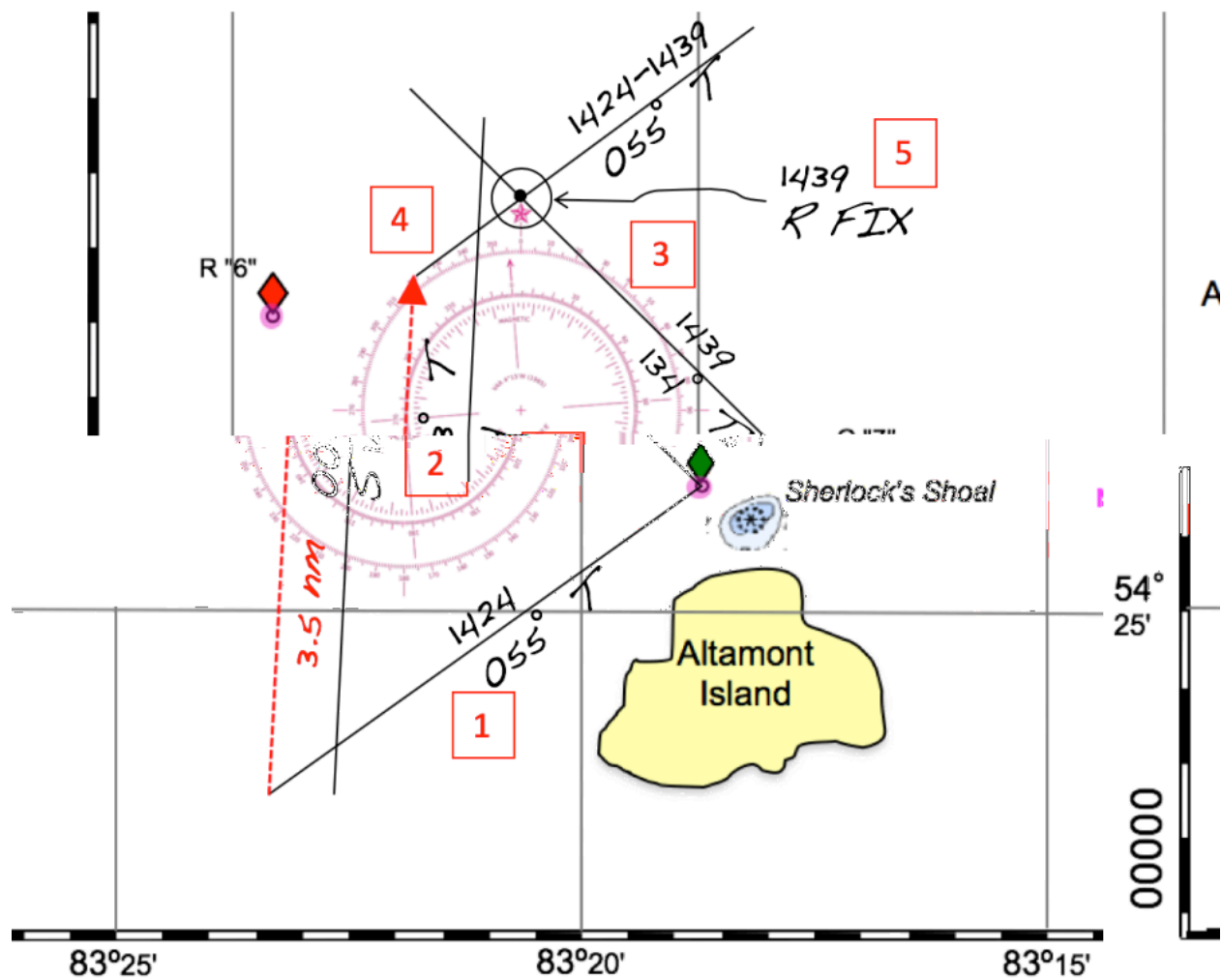


Figure 3.19. Taking a running fix.

60 D ST

1424-1439

3.4.12. Double the Angle on the Bow

double the angle on the bow

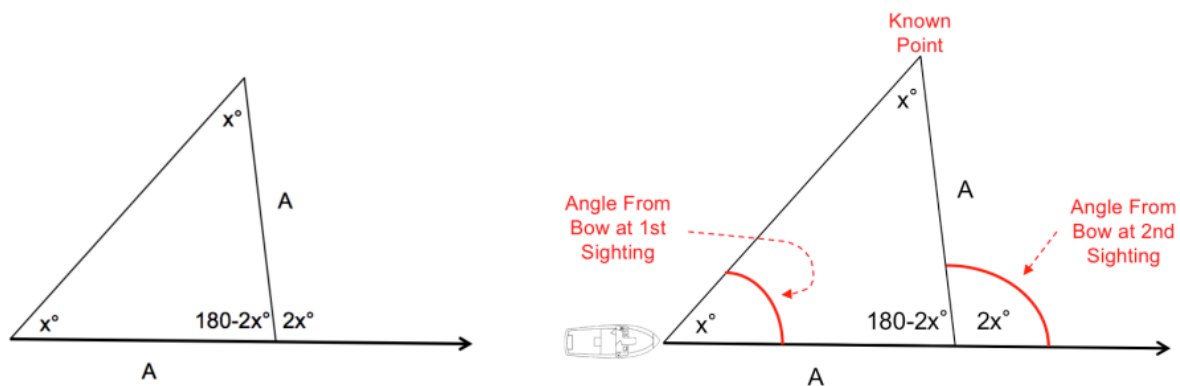


Figure 3.20. An isosceles triangle.

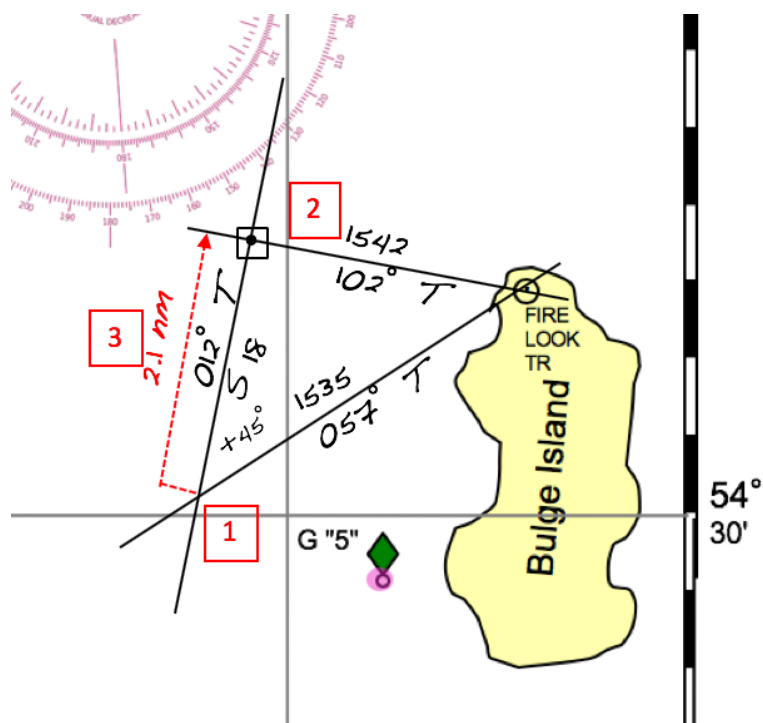


Figure 3.21. Double the angle on the bow.

3.4.13. Other Ways in Which to Get a Fix

range

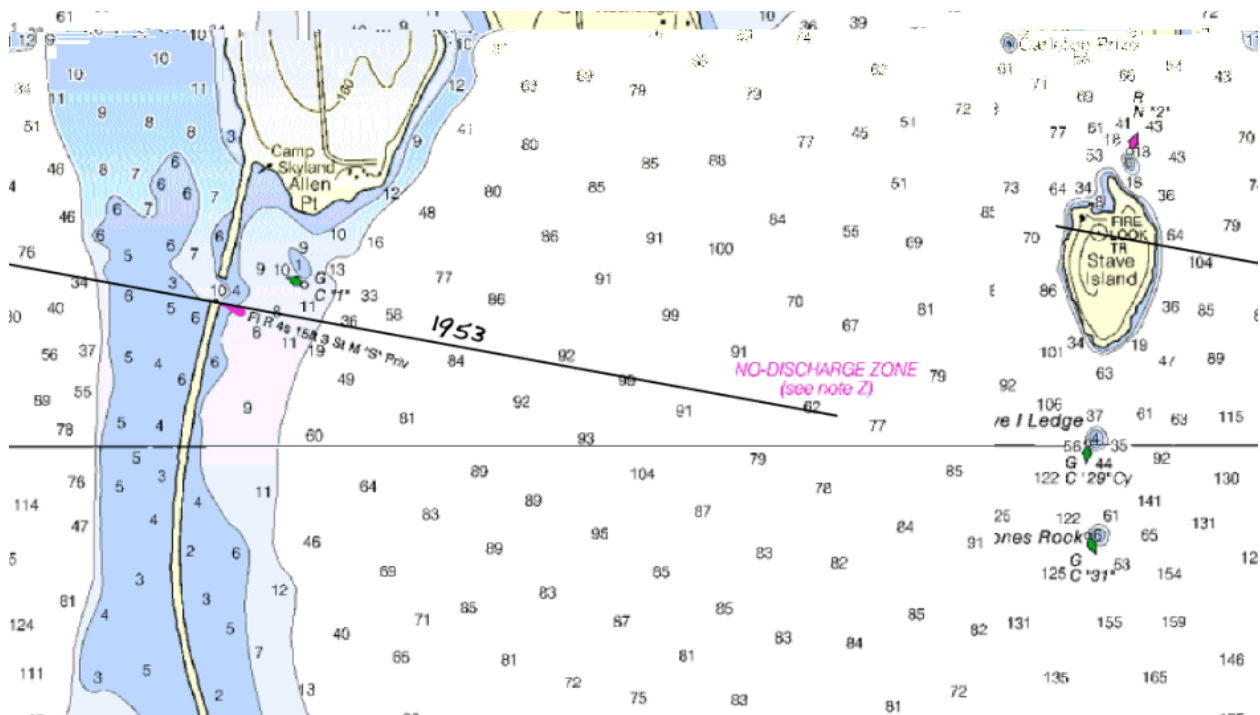


Figure 3.22. LOP from a range between a tower and light, from Chart 14782.

range

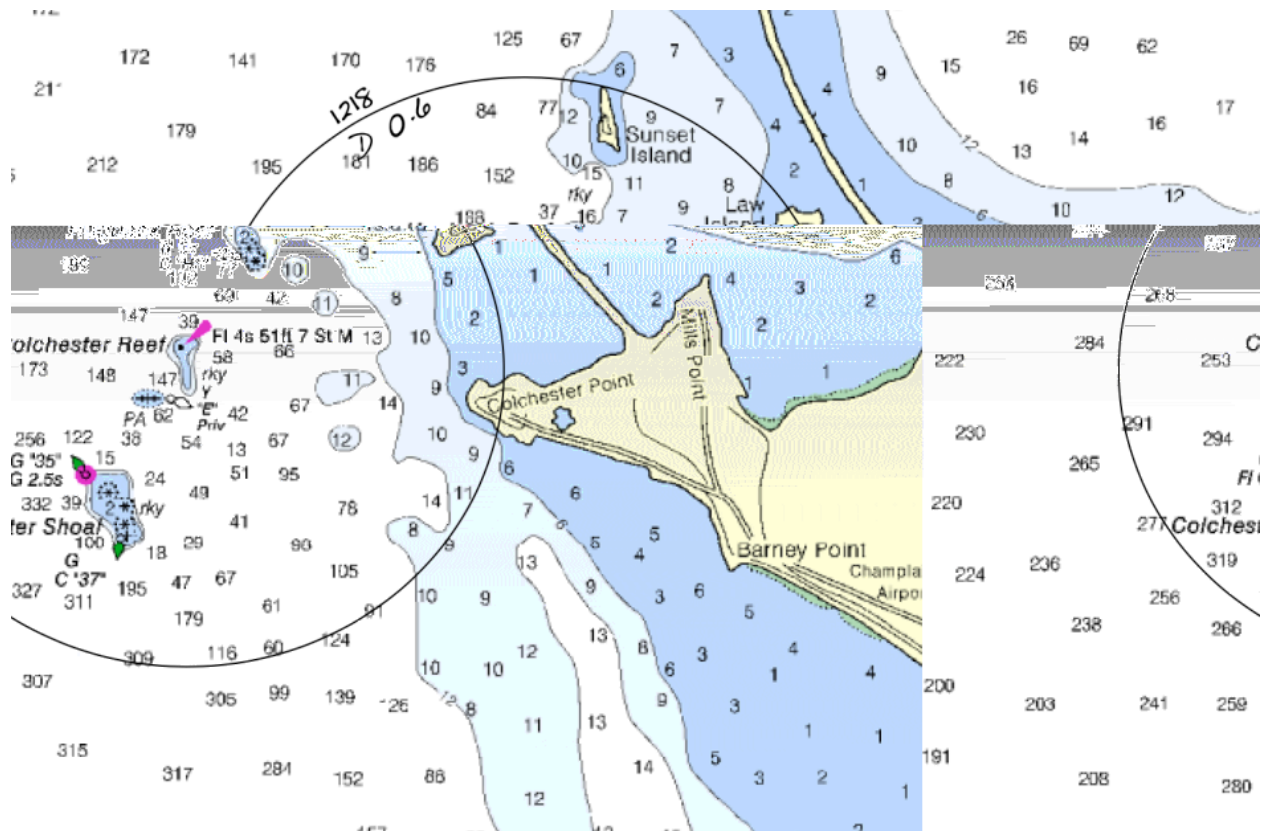


Figure 3.23. LOP based on a distance from a known object, from Chart 14782.

3.5. Voyage Planning

speed curve

fuel consumption table

BoatTEST.com

<i>Throttle (RPM)</i>	600	1,000	1,500	2,000	2,500	3,000	3,500	4,000
<i>STW (kn)</i>	3.1	5.3	7.9	9.8	16	19	23	26
<i>Fuel Burn Rate (GPH)</i>	0.3	0.7	1.9	4.9	8.4	14	22	31

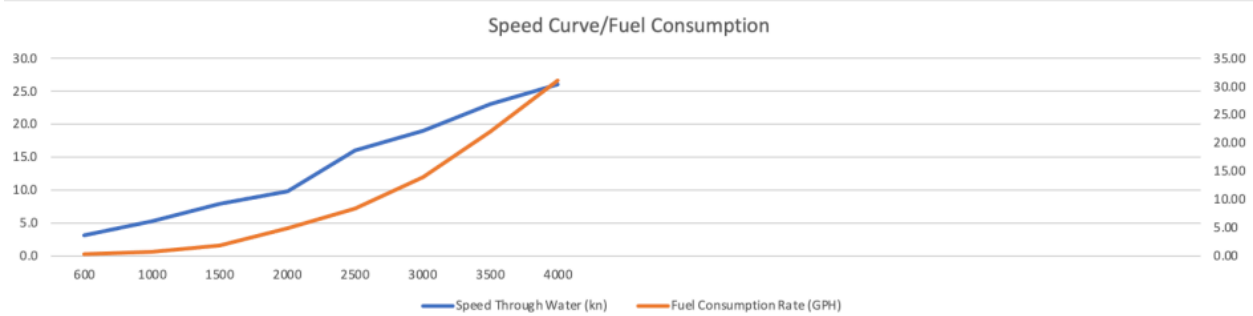


Figure 3.24. Sample Speed Curve/Fuel Consumption graph.

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- *Speed of advance (SOA):*
- *Fuel Efficiency (nm/gallon)*
- *Range (nm)*
- *Endurance (hours)*

⁴² An Excel version of this worksheet can be downloaded from <https://scuba.garykessler.net/library/PerformanceTable&Graph.xlsx>.

[illegible]

VOYAGE PLAN									
VESSEL: Mellow		DATE: 11/06/2020							
Description: Halifax Harbor Marina to Lady P (site 12)									
Starting Point	End Point	Distance (nm)	Throttle Setting (RPM)	Speed Through Water (kn)	Current (kn)	Fuel Burn Rate (GPH)	Speed of Advance (kn)	Fuel Used (gal)	Time (min)
Marina	BCN 40 Halifax R.	0.60	1500	7.9	0.0	1.90	7.9	0.14	4.6
BCN 40	Inlet channel, bouy 1	7.30	4000	26.0	0.5	31.00	26.5	8.54	16.5
Bouy 1	Disappearing Island	1.30	1500	7.9	1.0	1.90	8.9	0.28	8.8
Disappearing Is.	Mouth of inlet	1.60	4000	26.0	2.5	31.00	28.5	1.74	3.4
Inlet	Site 12	9.80	3000	19.0	0.0	14.00	19.0	7.22	30.9
Site 12	Mouth of inlet	9.80	3000	19.0	0.0	14.00	19.0	7.22	30.9
Inlet	Disappearing Island	1.60	4000	26.0	-3.0	31.00	23.0	2.16	4.2
Disappearing Is.	Inlet channel, bouy 1	1.30	1500	7.9	-2.0	1.90	5.9	0.42	13.2
Bouy 1	BCN 40 Halifax R.	7.30	4000	26.0	-0.5	31.00	25.5	8.87	17.2
BCN 40	Marina	0.60	1500	7.9	0.0	1.90	7.9	0.14	4.6
Total distance (nm):		40.6			Total fuel used (gallons):		36.7		
					Total time elapsed (minutes)		134		
					Total time elapsed (hour:min)		2:14		

<i>Throttle (RPM)</i>	<i>STW (kn)</i>	<i>Total Trip Time (hour:min)</i>	<i>Total Fuel Used (gallons)</i>

3.6. Summary

APPENDIX A: Expressing Latitude and Longitude

- *DMS*
-
- *GPS chart decimal*

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STEP 1:

STEP 2:

STEP 1:

STEP 2:

⁴³ One such site is <http://www.csgnetwork.com/gpscoordconv.html>. There are also apps available for Android and iOS platforms. A spreadsheet and Perl program for performing these calculations can also be found at <http://www.garykessler.net/software#latlong>.

EXAMPLE 1: Converting DMS to Decimal and GPS/Chart Notation

Ponce de Leon Inlet					
9515	- Lighted Bell Buoy 2	29-04-46.031N	FIR 2.5s	5	Red.
615		080-53-29.015W			

DMS Format: 29°04'46.031"N, 080°53'29.015"W

GPS/Chart Format: 29°04.767183', -080°53.483583'

04.767183

53.483583

Decimal Format: 29.079453°, -080.891393°

EXAMPLE 2: Converting Decimal to DMS and GPS/Chart Notation

Decimal Format: 29.554194°, 034.960732°

GPS/Chart Format: 29°33.25164', 034°57.64392'

DMS Format: 29°33'15.1"N, 034°57'38.63"E

EXAMPLE 3: Converting GPS/Chart Notation to DMS and Decimal

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GPS/Chart Format: 44°29.12000', -073°14.58000'

Decimal Format: 44.485333°, -073.243000°

DMS Format: 44°29'07.2"N, 073°14'34.8"W

⁴⁴ See Kessler, G.C. (2016, Fall). The Wrecks of Lake Champlain. *MAHSnews, The Newsletter of the Maritime Archaeological and Historical Society*, 27(2), 5-8. https://scuba.garykessler.net/library/WOLC_MAHs_2016%20Fall.pdf

APPENDIX B: Mercator and Polyconic Projections

projections

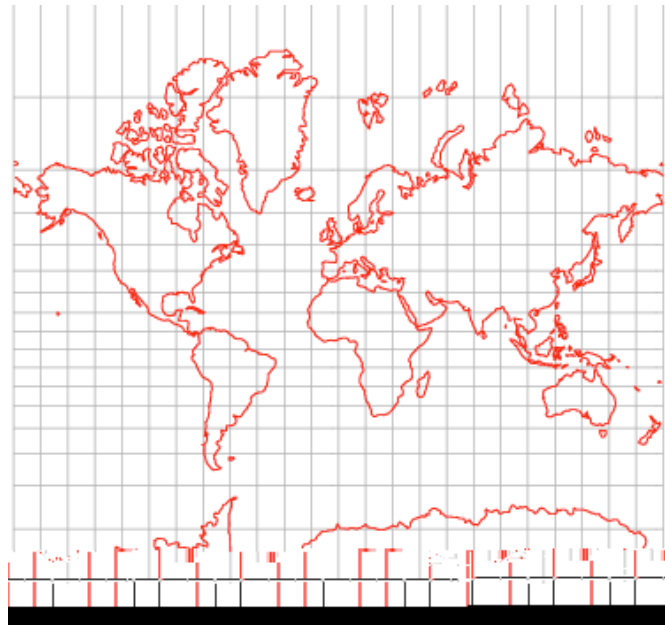


Figure B.1. Mercator projection.⁴⁵

46

rhumb line loxodrome

⁴⁵ <http://mathworld.wolfram.com/MercatorProjection.html>

⁴⁶ It is because lines of longitude are the same length that distance is always measured on the vertical scale of a nautical chart using Mercator projections. The horizontal scale shows degrees of longitude, which vary in distance depending upon how far north or south they are.

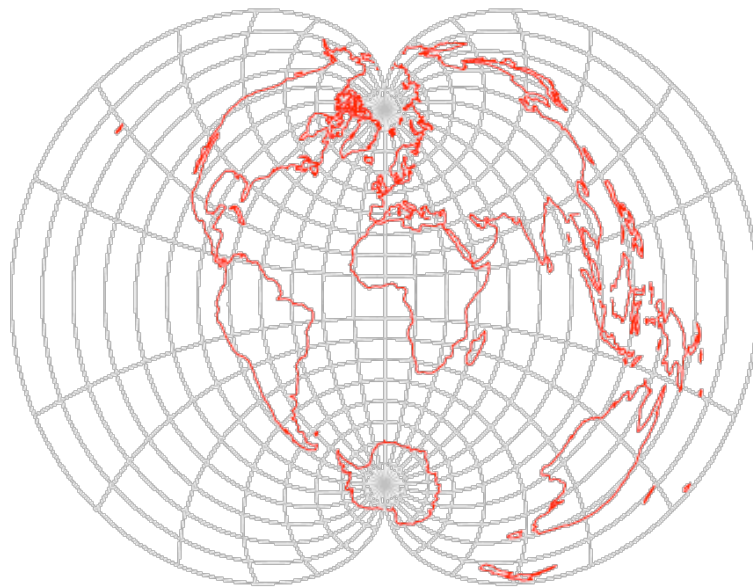
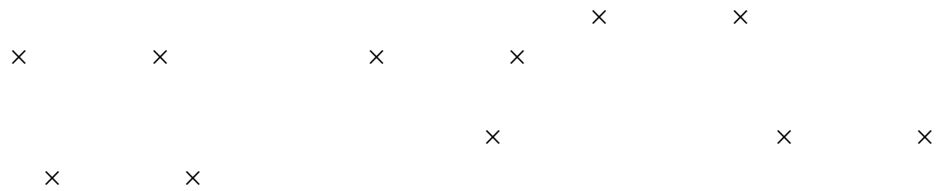


Figure B.2. Polyconic projection.⁴⁷

⁴⁷ <http://mathworld.wolfram.com/PolyconicProjection.html>

APPENDIX C: The Length of a Degree



<https://msi.nga.mil/msisitecontent/staticfiles/calculators/degree.html>
<http://www.csgnetwork.com/degreenllavcalc.html>

Latitude	One degree of latitude			One degree of longitude		
	sm	nm	km	sm	nm	km

Key: sm = statute miles; nm = nautical miles; km = kilometers

<https://www.nhc.noaa.gov/gccalc.shtml>

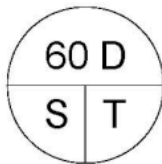
APPENDIX D: Navigators Quick Reference Card

<http://captnmike.com/2009/10/01/piloting-and-navigators-quick-reference/>

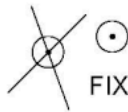
Navigators Quick Reference Card



025 or 025T – TRUE
 025M – MAGNETIC
 025C – COMPASS
 TIME = 4 DIGITS (24HR)
 COURSE = 3 DIGITS (000 (common) or 000.0)
 S 10 or S 10.5 = SPEED (0.1 KTS MOST APPS)
 RELATIVE BEARING + SHIPS HEADING = ACTUAL



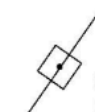
Distance
 Speed
 Time (Min)



FIX



DR



ESTIMATED
 POSITION

Speed (S): $S = (60 \times D) / T$

Speed in Knots or Nautical Miles per Hour

Time (T): $T = (60 \times D) / S$

Time is always in minutes

1 nm = 1 Minute of Latitude = 6000 ft = 2000 yards (Many Calculations)

60 nm = 1 Degree of Latitude

1 nm = 1852 meters = 6076 ft (International Treaties)

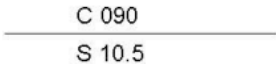
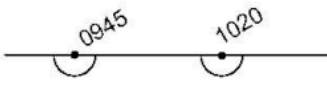
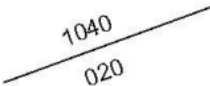
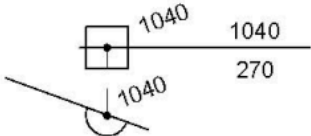
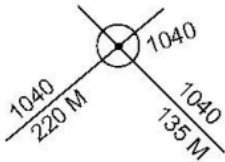
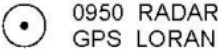
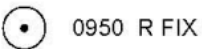
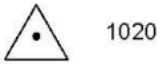
1 nm = 6100 ft (some calculations)

100 D = 1.24 M / H
 Always nautical miles (nm)

100 T / S = 1.24 M / H
 © 2005-2009 www.captnmike.com

100 D = 1.24 M / H
 Distance in

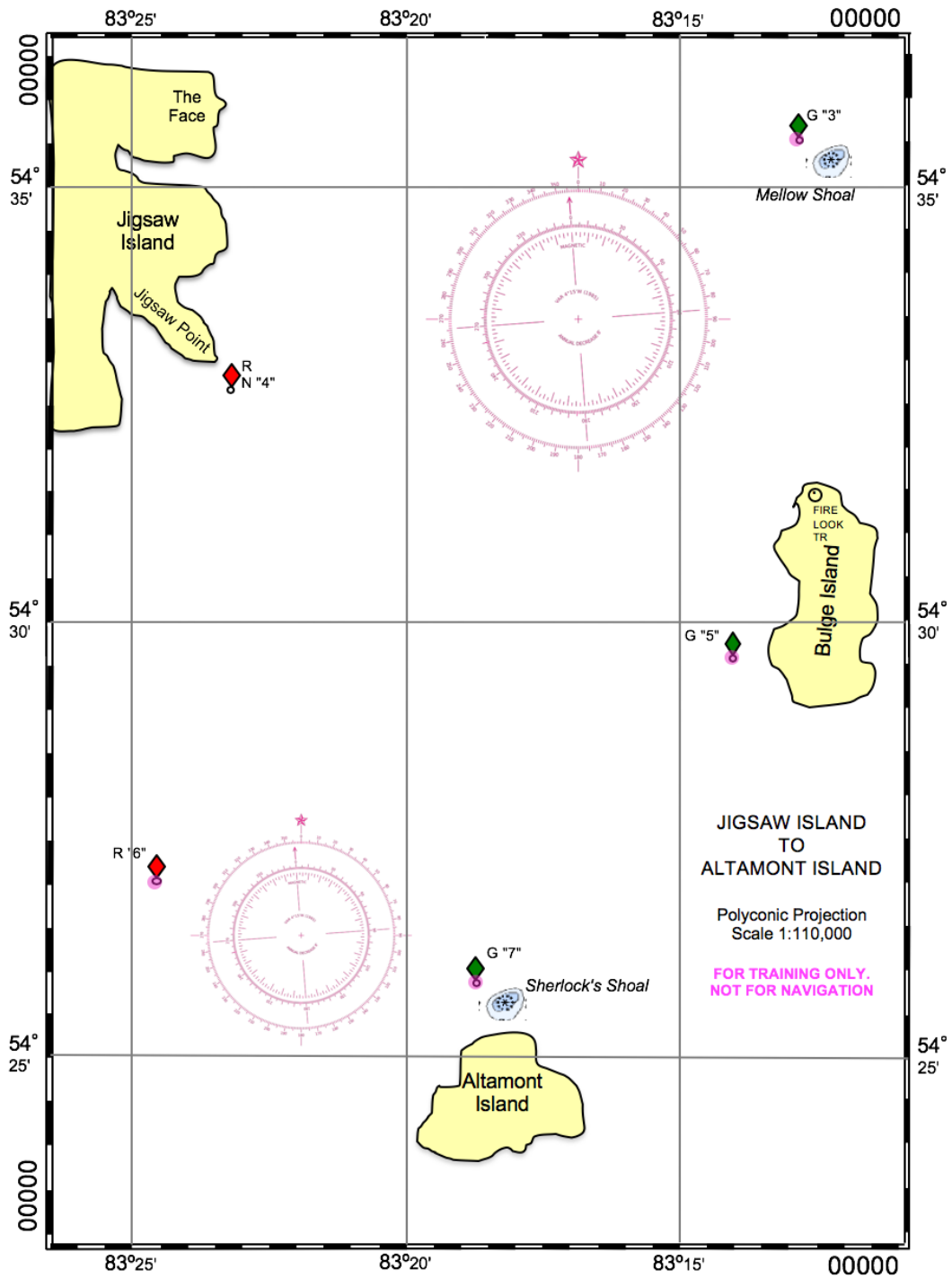
Navigators Quick Reference Card

ITEM	DIAGRAM	DESCRIPTION
DR Plot		Course (090 True) written above line, Speed (10.5 Knots) written below line
DR Position		Time (24 Hour) Written at angle to semicircle denoting DR position
LOP		Lightly draw line with Time (24 Hour) above LOP and True bearing beneath
Estimated Position		Square located where dashed perpendicular line from DR position touches LOP
Visual Fix		Circle where two or more LOP's cross. Time written parallel to LOP lines
Electronic Fix		Time and method (if relevant)
Running Fix		Circle containing the intersection of a given LOP and another LOP advanced (or retired) in time with RFIX and the time taken
Known Position		Triangle with Time written alongside

Summary of Common Navigation Drafting Symbols and Their Usage

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APPENDIX E: Sample Problem Chart⁴⁸



⁴⁸ This chart was created for example and training purposes only. The chart is wholly fictitious. A PDF version can be downloaded from <https://scuba.garykessler.net/library/SampleChart.pdf>.

Acronyms and Abbreviations

References and Further Reading

The American Practical Navigator: An Epitome of Navigation

Boat Navigation for the Rest of Us

Introduction to Electronic Chart Navigation

How to Read a Nautical Chart

Chapman Piloting & Seamanship

Dutton's Nautical Navigation

The Mariner's

Blog

Astronomical Algorithms

National Geophysical Data Center.

National Centers for Environmental Information.

Following

Seas Blog.

Longitude: The True Story of a Lone Genius Who Solved the Greatest Scientific Problem of His Time

GPS for Mariners

The Weekend Navigator

U.S. Aids to Navigation System

Navigation Rules – International and Inland Waters

Light List

United States Coast Pilot®

*Chart No. 1:
Symbols, Abbreviations and Terms used on Paper and Electronic Navigational Charts*

*International Code of Signals for Visual, Sound,
and Radio Communications (United States Edition).*

Get Your Captain's License: The Complete Study Guide

Acknowledgements

course made good

About the Author

Maritime Cybersecurity: A Guide for Leaders and Managers